



Bitcoin trade volume in decentralized markets: International evidence

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ABSTRACT

We analyze the determinants of Bitcoin (BTC) trade volume in decentralized exchanges (DEXs) and test the claim that BTC trades on these platforms are censorship-resistant. The study finds that overall economic freedom, particularly monetary freedom, correlates indirectly with BTC trade volumes, while capital restrictions on residents' transactions abroad correlate in two different directions. Purchase transactions inversely correlate with BTC volume in DEXs, while sales transactions correlate directly. These results suggest that BTC can be used to hedge against poor institutional frameworks, particularly against poor monetary governance, and as a vehicle for institutional hedging against repressive capital controls and institutional failures. The study's originality lies in its use of on-chain panel data on the volume of BTC transactions, which are country-specific and allow for comparing the impact of country-specific socio-institutional variables on BTC volumes.

1. Introduction

The “genesis” block of Bitcoin (BTC) blockchain transactions was mined on Saturday, January 3rd, 2009.¹ On this occasion, the first 50 BTCs were mined into existence. Despite its high volatility, BTC, one of the most elusive features of the internet era, is now the flagbearer of crypto-assets. Even though BTC and altcoins (i.e., crypto-assets leveraging other protocols, such as Ethereum, Solana, or Tron) are still emergent technologies and ecosystems compared to legacy financial structures, they significantly impact economic transactions and provide additional value to users and businesses alike. This is especially true in developing countries with poor institutional frameworks and financial development. Indeed, growth in crypto-adoption in the world is mainly driven by lower-middle-income countries, such as India, Nigeria, Ukraine, Indonesia, Pakistan, and Turkey, even though these countries usually implement severe restrictions (and sometimes outright ban) on holding and transacting crypto-assets (Chainalysis, 2024). At first glance, this may seem paradoxical. Yet, it merely illustrates the fact that implementing restrictive regulations to deter, or even ban, the use of decentralized technology is hardly efficient.

Nevertheless, BTC's price volatility raises significant challenges for portfolio diversification and institutional investors (Bouri et al., 2017a,

2017b; Platanakis and Urquhart, 2020). Indeed, whether BTC's historical valuation is a genuine reflection of its monetary soundness as an alternative to national fiat currencies or the indication of a speculative bubble is still under debate (Baur et al., 2018; Cheah and Fry, 2015; Cheung et al., 2015; Corbet et al., 2018a; Selgin, 2015). More recently, Hazlett and Luther (2020) argue that BTC is money (or near money) and that its valuation reflects its demand as a medium of exchange. Furthermore, studies question the nature of BTC as an asset: a hedge against other asset classes (Bouri et al., 2017a, 2017b), a diversification asset (Brière et al., 2015; Platanakis and Urquhart, 2020), a quasi-synthetic commodity money (Selgin, 2015), or a safe-haven asset (Smales, 2019; Urquhart and Zhang, 2019). The debate over the property of bitcoin hedge against inflation expectations is still open (Conlon et al., 2021; Liu and Valcarcel, 2024).

Interest in the determinants of BTC price and trade volume resulted in studies showing a diverse set of determinants from sentiments on social networks and searches on the internet (Aalborg et al., 2019; Naeem et al., 2021; Panagiotidis et al., 2018) to market forces (Ciaian et al., 2016). Recently, many articles have related BTC prices and trade volumes to more traditional macroeconomic variables and national uncertainties (Vo et al., 2021; Wu et al., 2021).

We propose an analysis of the determinants of BTC trade volumes

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¹ On-chain data of the Bitcoin blockchain, such as price, transactions volume, blocks, charts, mining, and network information, can be found on block explorers, such as <https://www.blockchain.com/explorer>.

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framed within institutional theory. This theoretical framework posits that institutions' structures, rules, and norms significantly influence economic behavior and decision-making processes (North, 1990). A country's regulatory environment, legal protections, and governance quality can impact BTC adoption and trade volumes. For instance, weak institutional frameworks and high levels of corruption may drive individuals and businesses toward decentralized financial systems, such as BTC, as alternatives to traditional financial systems (La Porta et al., 1999).

Institutional theory provides a lens through which we can analyze how differences in national governance and economic conditions affect the use and trade of Bitcoin. This approach aligns with recent studies highlighting the role of institutional quality in shaping financial markets and investor behavior (Acemoglu and Robinson, 2012; Djankov et al., 2007). By applying institutional theory, we aim to uncover the underlying institutional factors that influence BTC trade volumes, particularly on DEXs, which operate with fewer regulatory constraints than centralized exchanges (Crenshaw, 2021; Zetzsche et al., 2020).

For now, the literature has mainly concentrated on transactions of crypto-assets on centralized exchange (CEX) platforms, which roughly represent 85 % of the daily trade volume, the remaining 15 % being performed on decentralized exchange (DEX) platforms.² CEXs are increasingly falling under the scrutiny of national regulators like the Securities and Exchange Commission (SEC), the Financial Action Task Force (FATF), and the Financial Stability Board (FSB) in Western countries (Houben and Snyers, 2018; Sauce, 2022; Staff of the Global Legal Research Directorate, 2021). Thus, trading crypto-assets on CEXs naturally complies with regulations and is mainstream in its features (Auer and Claessens, 2018).³

On the other hand, decentralized finance (DeFi) is less exposed to regulators (Crenshaw, 2021; Zetzsche et al., 2020). The absence of regulation of decentralized finance makes DEXs the “wild west” of cryptocurrency trading (Aspris et al., 2021). However, a lack of regulatory foresight does not mean that regulation does not affect unregulated, decentralized exchange platforms. CEXs and DEXs are not impacted symmetrically by regulatory oversight. For instance, an outright ban precludes the registration of CEXs but incentivizes individuals to use DEXs to circumvent bans and other institutional factors. Regulation and institutional constraints directly impact CEXs but only indirectly impact DEXs. CEXs and DEXs are thus alternative ways to access crypto-related financial services. For the supporters of DEX transactions, the social and economic value of unregulated DEXs precisely lies in the possibility for individuals to circumvent regulation that curbs financial sovereignty.

DeFi leverages blockchain technology and smart contracts to provide a set of innovative and potentially disruptive financial functionalities and business models (Piñeiro-Chousa et al., 2022). Among the novelties are lending platforms, DEX platforms, synthetic assets, and yield farming (Angelis and Ribeiro da Silva, 2019; Chen and Bellavitis, 2020; Kimani et al., 2020). DEXs, contrary to CEXs, do not use order books but

are peer-to-peer trading platforms based on smart contracts (Cong and He, 2019).⁴ The main advantages of DEXs over CEXs are the significant number of token pairs that can be swapped (including tokens with low liquidity), fewer regulatory requirements, and lower transaction costs (Pereira et al., 2019). DEXs' trade volumes rapidly grow and exhibit efficient price dynamics (Makridis et al., 2023). These features offer users a more comprehensive offer of financial services, such as greater portfolio diversification when including crypto-assets and access to an ample choice of loans, whatever the regulation of the user's home country. While most DEXs are on the Ethereum protocol, others are exclusively dedicated to BTC trades. BTC DEXs are of particular significance because they allow users access to BTC, regardless of location and local regulatory and institutional frameworks, making BTC a “censorship-resistant” crypto-asset. Yet, BTC DEXs are largely absent from the literature dedicated to crypto-assets.

Our study aims to fill this gap, analyze the determinants of BTC trade volume in DEXs, and test the claim that BTC trades on these platforms are censorship-resistant (cf. Nabilou, 2021). For this purpose, we analyze the impact of a set of economic governance and capital control variables on the trade volume of BTC in the largest BTC DEXs, Localbitcoins, and Paxful.⁵ Overall economic freedom, particularly monetary freedom, correlates indirectly with BTC trade volumes, while capital restrictions on residents' transactions abroad correlate in two different directions. Purchase transactions inversely correlate with BTC volume in DEXs, while sales transactions correlate directly. These correlations suggest an incentive to trade in DEXs to evade regulation unless the latter enact outright controls over purchase transactions of non-domestic assets—access to CEXs or DEXs usually necessitates some access via traditional finance networks.

Moreover, controls over sales and issues of domestic assets abroad incentivize residents to look for an alternative asset, BTC. These characteristics of “governance hedging” suggest that BTC can be used to hedge against poor institutional frameworks, particularly against poor monetary governance. More than a medium of exchange or a speculative asset, bitcoin provides traders from countries with poor economic governance a way out for their capital to markets with better governance. Our results thus complement previous results on bitcoin as a portfolio diversification asset (Brière et al., 2015; Damianov and Elsayed, 2020) or as a resource for illegal activities, namely evading capital controls and poor institutional governance. We consider BTC a vehicle for institutional hedging against repressive capital controls and institutional failures. In the current context of the Russo-Ukrainian crisis, this alternative use for BTC could prove crucial for citizens trying to leave the conflict zone or a repressive regime while safely moving their capital through BTC DEXs (cf. Cheesman, 2022).

Our research extends the existing literature by focusing on the impact of economic governance and capital restrictions on Bitcoin trade volumes in DEXs. We examine how overall economic freedom, particularly monetary freedom and capital restrictions, correlate with BTC trade volumes. This analysis is crucial for understanding BTC's role as a vehicle for institutional hedging against poor monetary governance and repressive capital controls. Our findings contribute to the ongoing debate on BTC's functionality as a medium of exchange, a speculative asset, and a tool for circumventing institutional failures. By leveraging institutional theory, we provide a comprehensive framework to analyze the dynamics of BTC trade volumes in the context of varying institutional and economic conditions.

² See <https://www.theblockcrypto.com/data/decentralized-finance/dex-non-custodial/dex-to-cex-spot-trade-volume>.

³ However, despite the increasing scrutiny of regulators, misbehavior, mismanagement, and frauds are still frequent in this infant industry. The recent turmoil of the FTX CEX is an illustration of the problem. FTX—and over a hundred affiliates—filed for chapter 11 on November 11th, 2022 because of a massive liquidity crisis. Before its collapse, FTX was the third largest CEX, with over one million users. Its collapse was a major event for the crypto-industry, but also for traditional finance. For instance, the crypto-friendly Silvergate bank declared bankruptcy on March 10th, 2023, after winding down operations and voluntarily liquidating its assets.

⁴ Smart contracts are digital contracts stored on a blockchain that self-execute contractual agreements when the terms of the contract are met (Szabo, 1996). The terms of the contract are transparent, immutable, objective (quantified or binary), determined peer-to-peer between the trading partners.

⁵ Due to unfavorable market conditions, Localbitcoins recently announced its closure in March 2023. However, it has no impact on the reliability of the data used or on the consistency of our results.

Another originality of our study is that we use on-chain panel data on the volume of BTC transactions on the Localbitcoins and the Paxful DEXs. These platforms' data are country-specific, so we can compare the impact of country-specific socio-institutional variables on BTC volumes. Country-undifferentiated data would preclude such analysis. Previous studies on BTC and its relation to socio-institutional factors use app downloads by country data (Luther and Salter, 2017), survey-based data on BTC adoption (Alnasaa et al., 2022), or price premia data (Johnson, 2020). Although those data provide relevant information on the penetration of crypto-assets and their volatility, they do not reveal much about the country dynamics of using crypto-assets over time. Moreover, on-chain data are the most reliable, transparent, and auditable data on blockchain-based transactions since they are the only data publicly recorded on the blockchain. We obtain similar results to those previous studies regarding macroeconomic control variables while using more reliable data and identifying overall and monetary freedom, as well as capital restrictions as significant explanatory variables of transaction volume in the Localbitcoins and Paxful DEXs. Given the countries where trade volume varies the most, our results suggest that BTC can be used as an instrument of capital mobility between countries and, hence, a vehicle of institutional hedging. Consequently, calls for regulation should be nuanced regarding which crypto-users the regulation will target. Under certain conditions, crypto-assets can prove helpful for citizens living in countries with poor institutional frameworks and stringent capital controls, refugees, and citizens emigrating from repressive regimes.

2. Literature review

2.1. Institutions and uncertainty

An emerging trend in the literature focuses on the quality of institutions, economic policy, economic policy uncertainty (EPU), and regulation as determinants of crypto-asset returns (Gozgor et al., 2019), in line with institutional theory (La Porta et al., 1999; North, 1990). For instance, Johnson (2020) tests bitcoin premiums across seventeen countries, finding that BTC premium depends on corruption and economic freedom. This suggests that individuals are ready to pay a premium when corruption increases, and economic freedom decreases. This result aligns with the assumption that BTC is a hedge against institutional failures (such as corruption or poor banking and financial systems), a proposition that BTC enthusiasts often put forward to illustrate the "censorship-resistance" property of bitcoin.

Similarly, Alnasaa et al. (2022) explore the correlation between the adoption of crypto-assets and indicators of corruption, capital control, and average consumer price inflation rates. They find that using crypto-assets is significantly and positively associated with a higher perception of corruption and more intensive capital controls, with crypto-assets being used to circumvent capital controls (Graf von Luckner et al., 2021).

Several studies confirm some hedging features of bitcoin relative to economic policy uncertainty (EPU). Demir et al. (2018) find an inverse relationship between EPU and BTC returns under stable periods of uncertainty, while this relationship turns positive at extreme times of uncertainty. Fang et al. (2019) find that EPU levels impact BTC's ability as a hedge differently against equities and bonds. Both Wang et al. (2019) and Wu et al. (2019) confirm BTC as a strong hedge against extreme EPU variations, but not during calmer times. Matkovskyy et al. (2020) and Umar et al. (2021) obtain similar results. Finally, Su et al. (2020) find that BTC is an efficient hedge against EPU and geopolitical shocks.

Instead of focusing solely on EPU, Aysan et al. (2019) focus on geopolitical risks and observe that geopolitical risk has a significant predictive power on returns and volatility of bitcoin. Accordingly, BTC can be considered a hedging tool against global geopolitical risk. Colon et al. (2021) mitigate these results, arguing that the crypto-asset market can be considered a strong hedge against geopolitical risks in most cases

but a weak hedge and safe haven against economic policy uncertainty during a bull market. For their part, Al Mamun et al. (2020) analyze the impact of geopolitical risk and EPU on the correlation between bitcoin and other asset classes and conclude that bitcoin investors can hedge their portfolio only with gold but not with other financial asset classes.

Some studies draw normative and political implications from the relationship between BTC prices and economic policy uncertainty. For instance, considering BTC is an efficient hedge against EPU and geopolitical risk shocks, Su et al. (2020) argue that governments should show interest in promoting BTC and blockchain technology. They further argue that their promotion must be accompanied by more substantial supervision and regulation to promote the healthy development of bitcoin markets. Furthermore, Luther (2022a), using a difference-in-difference approach, shows that the de-platforming of Pornhub by Visa, Mastercard, and Discover helped increase the demand for BTC as users rushed toward this asset as a means of payment to deal with Pornhub.

Contrary to the normative conclusion that tighter regulation is desirable, other studies question the desirability and efficiency of regulation and insist on the detrimental consequences of regulation on crypto markets. Relying on the literature on entrepreneurial activity, Luther (2022b) argues that regulatory ambiguity and jurisdictional redundancy prevent the sound development of the crypto-industry. Using data on 120 regulatory events, Shanaev et al. (2020) analyze the implications of regulation on 300 crypto-assets. They find that tighter regulation decreases crypto-asset prices. The rationale is that consumer utility losses outperform the decrease in risks usually associated with tighter regulation. This argument suggests that crypto markets efficiently integrate information on regulation and that tighter regulation is undesirable. Sauce (2022) investigates whether applying the latest regulatory guidance of the Financial Action Task Force (FATF) may engender unintended detrimental consequences on the organization and efficiency of cryptomarkets. It is argued that tighter regulation incites some investors to flee to unregulated, riskier, and less liquid DEX platforms. This flight leads to resource misallocation and a two-tier industry. On one side are compliant CEXs, and unregulated DEXs on the other. These unintended consequences are at odds with the initial purpose of regulators to devise a technology-neutral regulation to improve overall market transparency and efficiency.

2.2. Crypto-assets as portfolio diversification assets

The most significant part of the literature on crypto-assets focuses on the financial dimension of crypto-assets, such as price dynamics (Ciaian et al., 2016), returns and performance (Bianchi and Babiak, 2022; Liu et al., 2022), volatility (Hafner, 2020), anomalies (Dong et al., 2022), liquidity risks (Brauneis et al., 2021; Zhang and Li, 2021), and speculation bubbles in crypto-asset markets (Bariviera and Merediz-Solà, 2021; Fakhfekh and Jeribi, 2020).

The empirical results on the efficiency of cryptomarkets are somewhat mixed; for now, there is no consistent accumulation of knowledge around a set of stylized facts and empirical results (Corbet et al., 2019; Urquhart and Zhang, 2019). Furthermore, it appears that the results heavily depend on the type, origin (off-chain or on-chain data from several CEXs), and period of data used. This can be explained by the fact that crypto-markets are still immature and in the process of organization, with higher levels of liquidity and aggregate market capitalization (from US\$18 billion in January 2017 to an all-time high of US\$ 3 trillion in November 2021), and the entry of institutional investors.⁶

Previous studies showed a decoupling and statistical decorrelation between crypto-markets and classical asset classes (Brière et al., 2015; Corbet et al., 2018b). These results suggest that crypto-assets may offer diversification benefits and that crypto-assets and traditional financial

⁶ Data from <https://www.coingecko.com/fr/global-charts>.

assets are owned and managed for different purposes and in different investment horizons. Baur et al. (2018) find that bitcoin is uncorrelated with traditional asset classes (bonds, stocks, and commodities) in regular times and bearish markets and conclude that investors use bitcoin mainly as a speculative investment in a strategy of portfolio diversification rather than as a medium of exchange. Similarly, Giudici and Abu-Hashish (2019) show that bitcoin prices are unrelated to classical financial market prices, confirming that crypto-assets have diversification benefits.

2.3. Market efficiency

In their analysis of market efficiency and volatility persistence in twelve crypto-assets during pre- and post-crash periods, Yaya et al. (2021) found that markets for bitcoins and other altcoins can be considered efficient though volatile. Following Urquhart (2016), who argued that the BTC market is moving toward informational efficiency, Vidal-Tomás (2022) argues that the BTC market is more efficient over time. Through their crypto-market performance analysis, Mnif et al. (2020) found that market efficiency increased during the pandemic.

On the other hand, Hu et al. (2022) found evidence of order-based price manipulation during the bitcoin bubble of late 2017. This finding is based on the definition of price manipulation as a violator's intent to pursue a scheme that undermines economic efficiency by making prices less accurate as signals for efficient resource allocation and making markets less liquid for risk transfer (Kyle and Viswanathan, 2008). In the same way, using trading data from 2013 on the now long-time defunct Mt. Gox exchange platform, Gandal et al. (2018) identify the role of suspicious bitcoin trades on price and trading volume.

Focusing on arbitrage and price deviations, Makarov and Schoar (2020) identify several new stylized facts about the price formation across crypto-asset markets, such as large and recurring deviations in bitcoin prices across exchanges. These differences in BTC valuation are more significant across countries and regions than within the same country, even between countries with liquid crypto-markets, such as the US, Japan, South Korea, and Europe. To explain the deviation from the law of one price, they conjecture that price deviations may reflect tighter capital controls or weaker financial institutions in countries other than the United States and European ones, arguing that controls on capital flows and barriers to arbitrage between regions reduce the efficiency of arbitrage. The rationale is that investors who want to evade strict capital controls are ready to buy crypto-assets at a higher price. Chen et al. (2022) extend the literature on the impact of capital control on price discrepancies in crypto-markets across countries by arguing that investors perceive crypto-assets as an alternative hedging investment under uncertainty, specifically during the Covid-19 pandemic and the implementation of strict lockdown regulation.

In this study, we follow Makarov and Schoar (2020), who hypothesize a long-lasting relationship between crypto-markets and the institutional governance framework, and Alnasaa et al. (2022), who analyze the empirical association between crypto-assets, perceived corruption, and capital controls, and works on the impact of economic policy uncertainty and BTC returns. However, while these studies concentrate on CEXs, we analyze the effects of economic governance, capital restrictions, and traditional macroeconomic variables on BTC weekly transactions on DEXs, namely Localbitcoins and Paxful. BTC DEX trading is a hedge against poor institutional frameworks, especially monetary freedom and capital controls. Therefore, based on a new set of data, we test the following hypotheses:

H1. : poor institutional quality positively impacts BTC trade volume in DEXs.

H2. : low economic freedom positively impacts BTC trade volume in DEXs.

H3. : stricter capital controls positively impact BTC trade volume in

DEXs.

3. Data and methodology

3.1. BTC trading volume

For the present study, we use the weekly volumes of BTC transactions of the two largest decentralized exchange platforms of bitcoin (Localbitcoins and Paxful) as endogenous variables. We restrict our analysis to these two DEXs for two reasons. First, they are BTC's most prominent decentralized exchange platforms, operating worldwide and exhibiting significant trade volumes. Other DEXs exhibit much smaller trade volumes and operate at a lower scale. Second, data about smaller DEXs are scarcer and less reliable. These limitations would prevent us from making international comparisons.

We use public and auditable "on-chain data," i.e., data recorded on the Bitcoin blockchain collected on the coin.dance website. On-chain data are specific to the blockchain technology. They are the most transparent and reliable crypto-transaction data (G. Kim et al., 2022). On the other hand, "off-chain" data are provided by external data service providers, whose methods of collect and treatment may be opaque and hardly comparable. The further advantages of this dataset over country-undifferentiated data on BTC transactions are twofold. First, it provides country-specific transaction data for the Localbitcoins (47 countries) and Paxful (14 countries) platforms.⁷ This country-specificity allows us to compare the impact of country-specific economic and socio-economic variables on BTC transactions. This analysis would be impossible if we used country-undifferentiated data. Accordingly, we add an international dimension that numerous previous studies fail to consider. Second, Localbitcoins and Paxful are DEX platforms, meaning that individuals can perform BTC transactions whatever the local regulation, even in countries that strictly prohibit the trade and holding of crypto-assets (such as Iran) and without the Know Your Customer (KYC) and Anti-Money Laundering (AML) requirements of centralized exchange platforms.

Furthermore, the platforms provide a wide range of means of payment (credit card, wire transfer, cash, gift cards) that allow even unbanked people to perform BTC transactions, even though the more "exotic" means of payment remain marginal. These features correspond to the initial vision of BTC, a "censorship-resistant" tool to achieve financial inclusion. Accordingly, considering BTC transactions on decentralized exchange platforms permits us to consider transactions that cannot be accounted for in data from centralized exchange platforms, as most studies usually do. The main drawbacks of on-chain data over off-chain data are that the validation of transactions is slower (the average time for mining a block of Bitcoin transactions is settled at 10 min) and more expensive (with mining activities). However, since we do not consider scalability and mining issues, these limitations are not significant for our purpose.

We manually collected the total number of BTCs traded weekly against 47 countries over the 2013–2019 period on the Localbitcoins and Paxful platforms.⁸ Our first dependent variable is the total volume by year and by country. We use the following notation: subindex i for the country, j for the year, and k for the week, with $i = 1, \dots, 47$ and $j = 2013, \dots, 2019$ and $k = 1, \dots, 52$. Then, our first dependent variable is:

$$Y1_{ij} = \sum_{k=1}^{52} VOLUME_{ij,k}^{LBC}$$

where $VOLUME_{ij,k}^{LBC}$ is the volume recorded on the DEX Localbitcoins for

⁷ See Table 1 for the list of countries covered by each platform.

⁸ See <https://coin.dance/volume/localbitcoins> for Localbitcoins and <https://coin.dance/volume/paxful> for Paxful. These data are also used in Wüstenfeld and Geldner (2022).

Table 1

Summary statistics on the dependent variables by country.

Variables	Y1	Y2	Y3	Y4
Definitions	Total Localbitcoins volume by country-year	Total Localbitcoins and Paxful volume by country-year	Localbitcoins volume scaled by outstanding BTC	Localbitcoins and Paxful volume scaled by outstanding BTC
Countries				
Argentina*	2.570	2.573	0.172	0.172
Australia*	37.018	37.764	2.594	2.638
Brazil	3.013	3.013	0.192	0.192
Canada*	10.713	12.531	0.727	0.836
Chile	1.157	1.157	0.072	0.072
China*	36.212	37.703	2.206	2.292
Colombia	7.091	7.091	0.419	0.419
Croatia	0.611	0.611	0.041	0.041
Czech Republic	0.799	0.799	0.056	0.056
Denmark	0.330	0.330	0.021	0.021
Dominican Republic	0.373	0.373	0.022	0.022
Egypt	0.074	0.074	0.004	0.004
Hong Kong*	3.839	3.839	0.251	0.251
Hungary	0.189	0.189	0.013	0.013
India*	8.429	8.827	0.532	0.555
Indonesia	0.241	0.241	0.014	0.014
Iran	0.778	0.778	0.048	0.048
Japan*	0.223	0.344	0.015	0.022
Kazakhstan	0.285	0.285	0.017	0.017
Kenya*	2.276	2.472	0.139	0.150
Malaysia*	5.472	5.494	0.347	0.348
Mexico	3.595	3.595	0.239	0.239
Morocco	0.458	0.458	0.027	0.027
New Zealand*	3.762	3.871	0.248	0.254
Nigeria	16.366	16.366	0.972	0.972
Norway	3.579	3.579	0.237	0.237
Pakistan	2.443	2.443	0.149	0.149
Peru	2.572	2.572	0.151	0.151
Philippines*	1.911	1.918	0.128	0.128
Poland	1.357	1.357	0.091	0.091
Romania	3.084	3.084	0.203	0.203
Russia	108.619	108.619	6.760	6.760
Saudi Arabia	1.182	1.182	0.070	0.070
Singapore	1.617	1.617	0.104	0.104
South Africa	16.547	16.547	1.082	1.082
South Korea	0.232	0.232	0.015	0.015
Sweden*	10.139	10.154	0.669	0.670
Switzerland	1.908	1.908	0.134	0.134
Tanzania	0.223	0.223	0.013	0.013
Thailand	8.714	8.714	0.589	0.589
Turkey	0.855	0.855	0.053	0.053
United States*	299.221	363.284	20.591	24.436
Ukraine	2.441	2.441	0.146	0.146
United Arab Emirates*	1.976	1.980	0.123	0.123
United Kingdom	113.507	114.530	8.001	8.060
Venezuela	16.836	16.836	0.986	0.986
Vietnam	0.266	0.266	0.016	0.016

The first column presents the list of countries covered by Localbitcoins. * indicates that the country is also covered by Paxful. Column 2 to 4 report the average values of Y1, Y2, Y3, and Y4. Y1 and Y3 are in thousands of BTC. Y2 and Y4 are multiplied by one thousand.

country i , year j , and week k . We thus obtain 329 country-year observations. Our second dependent variable combines the BTC volume from Localbitcoin and Paxful for the same country i , year j , and week k .

$$Y2_{ij} = \sum_{k=1}^{52} (VOLUME_{ij,k}^{LBC} + VOLUME_{ij,k}^{PXL})$$

For robustness checking, we create two other dependent variables. Since the BTC outstanding amount tends to increase progressively due to mining activities, we propose a third dependent variable where annual volumes are scaled by $OUSTANDING_BTC_j$, the average number of BTC mined in the year j . Formally,

$$Y3_{ij} = \frac{\sum_{k=1}^{52} VOLUME_{ij,k}^{LBC}}{\frac{1}{52} \sum_{k=1}^{52} OUSTANDING_BTC_{k,j}}$$

Our fourth dependent variable scales the previously combined volumes by $OUSTANDING_BTC_j$:

$$Y4_{ij} = \frac{\sum_{k=1}^{52} (VOLUME_{ij,k}^{LBC} + VOLUME_{ij,k}^{PXL})}{\frac{1}{52} \sum_{k=1}^{52} OUSTANDING_BTC_{k,j}}$$

Descriptive Statistics on the dependent variables are reported in Table 1.

Table 2

Summary statistics on institutional scores and indices by countries.

Countries	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Argentina	47.729	25.857	33.114	42.867	65.471	50.600	47.333	56.100	45.114	55.200	68.529	41.429	41.429
Australia	81.300	85.643	81.257	90.933	63.786	61.357	85.033	91.471	81.343	85.029	86.486	80.714	90.000
Brazil	54.843	51.871	36.257	52.300	69.771	53.429	12.133	57.086	51.800	70.257	69.271	51.429	54.286
Canada	78.657	88.971	83.029	75.767	78.614	49.529	81.533	85.343	76.029	76.971	87.886	79.286	80.000
Chile	77.286	79.971	68.900	61.133	76.900	82.629	92.267	71.786	66.357	83.757	85.186	86.429	70.000
China	54.671	32.129	40.714	67.100	70.043	76.671	84.800	52.714	62.643	72.114	72.600	25.714	25.714
Colombia	69.814	54.814	35.243	31.967	78.829	74.871	83.733	80.314	78.671	77.671	78.914	77.857	70.000
Croatia	60.586	50.343	42.800	52.067	69.957	38.843	65.767	59.757	42.471	79.557	87.186	76.429	60.000
Czech Republic	72.857	72.586	49.629	53.800	82.300	45.886	95.267	68.971	80.386	82.700	87.214	78.571	80.000
Denmark	76.029	89.671	89.357	76.633	39.829	5.886	96.267	95.200	87.914	84.000	87.214	89.286	82.857
Dominican Republic	61.467	40.567	28.100	22.167	84.350	90.117	90.167	54.017	56.217	76.383	76.333	74.167	40.000
Egypt	53.914	28.586	31.529	52.367	85.443	67.014	1.933	66.271	49.771	64.800	71.243	53.571	42.857
Hong Kong	89.686	91.357	80.314	81.200	92.986	89.929	100.000	97.500	90.614	83.086	90.714	90.000	90.000
Hungary	66.371	57.657	45.114	51.367	79.243	27.929	82.233	70.057	65.629	83.614	87.214	76.429	70.000
India	54.857	54.014	39.400	53.433	78.600	77.786	12.967	47.457	52.757	70.314	68.886	37.143	40.000
Indonesia	60.686	38.543	35.857	46.000	83.500	89.757	89.200	54.857	49.300	76.300	77.971	39.286	58.571
Iran	46.350	21.400	28.767	37.533	81.000	90.950	92.033	61.650	50.900	53.667	50.167	0.833	10.000
Japan	72.086	82.786	78.729	71.833	68.529	49.543	38.167	81.771	81.414	85.757	82.043	68.571	55.714
Kazakhstan	65.683	42.733	33.933	56.900	93.067	85.400	75.733	73.850	85.317	72.667	78.117	43.333	50.000
Kenya	55.629	38.114	25.614	44.533	78.486	76.314	14.100	53.086	62.257	73.643	66.643	53.571	50.000
Malaysia	71.471	67.600	50.186	66.900	85.086	77.229	79.867	87.000	74.471	82.229	80.771	56.429	54.286
Mexico	65.500	53.686	30.271	37.533	77.300	77.957	73.267	72.343	58.914	77.714	82.914	71.429	60.000
Morocco	60.800	46.571	37.271	44.433	71.357	66.800	60.833	70.771	33.171	81.643	75.771	68.571	64.286
New Zealand	82.657	95.171	93.329	86.800	70.914	42.986	98.233	93.886	87.757	87.557	87.829	80.000	80.000
Nigeria	56.486	32.829	20.829	35.700	84.643	86.829	78.767	50.014	74.700	70.214	63.786	41.429	40.000
Norway	72.186	88.457	89.186	83.500	53.757	37.300	97.833	90.643	49.014	76.943	87.771	74.286	60.000
Pakistan	54.817	33.983	28.017	36.100	79.200	87.283	44.667	61.467	41.950	72.717	65.550	51.667	40.000
Peru	67.983	48.550	36.200	31.900	79.617	86.750	93.867	68.750	62.733	83.533	86.933	72.500	60.000
Philippines	62.571	37.557	32.871	37.233	78.757	89.829	97.333	59.686	55.529	76.800	76.871	58.571	57.143
Poland	67.929	61.414	55.286	52.867	76.643	46.186	81.333	67.214	61.571	81.971	87.214	72.857	70.000
Romania	67.214	49.514	40.771	56.700	87.671	64.357	90.433	67.357	65.171	79.900	87.214	76.429	50.000
Russia	54.271	34.100	30.571	45.500	85.400	59.457	89.233	73.986	54.314	63.729	75.971	27.143	30.000
Saudi Arabia	61.850	48.350	47.383	62.633	99.717	57.633	34.933	70.517	68.933	71.383	76.767	40.833	50.000
Singapore	88.771	93.271	89.729	91.600	90.857	91.114	80.233	94.657	92.843	83.400	90.686	83.571	80.000
South Africa	61.771	56.300	43.043	54.967	67.657	68.643	69.067	69.043	58.343	75.171	75.843	47.143	55.714
South Korea	72.157	74.500	55.100	60.367	71.871	69.900	97.067	91.400	53.043	81.657	76.100	69.286	74.286
Sweden	73.871	90.100	89.943	84.800	43.057	21.157	95.367	90.000	53.543	84.043	87.214	87.143	80.000
Switzerland	81.314	88.057	85.457	80.567	70.014	65.514	96.000	77.057	77.329	85.757	89.629	84.286	84.286
Tanzania	58.940	32.440	31.640	34.967	79.940	87.460	80.267	47.840	63.680	69.540	72.180	57.000	50.000
Thailand	65.043	46.229	36.057	44.300	80.686	83.929	96.400	74.714	64.257	71.614	78.871	48.571	62.857
Turkey	64.043	50.971	43.557	52.267	76.057	61.786	93.833	65.114	49.557	71.929	82.329	72.143	60.000
US	75.814	80.700	73.914	76.867	67.486	53.129	53.733	86.300	93.486	77.043	86.800	75.714	72.857
Ukraine	48.800	33.043	26.243	27.867	79.314	36.514	75.467	59.200	49.157	65.900	83.457	24.286	30.000
United Arab Emirates	74.229	64.971	71.957	85.167	98.286	76.743	95.700	77.657	81.157	81.871	83.214	39.286	55.714
United Kingdom	76.457	91.186	78.500	90.900	62.343	37.286	54.167	91.014	73.257	78.300	87.214	90.000	80.000
Venezuela	31.214	5.657	14.500	12.400	74.357	54.000	17.067	40.700	28.943	27.200	60.986	1.429	14.286
Vietnam	52.883	31.817	29.650	36.200	79.067	74.383	29.700	61.617	63.200	70.217	80.217	22.500	36.667

This table reports average Heritage indexes by country. High (low) scores mean an important (poor) entrenchment of the institutional factor. (1): Overall Score. (2): Property Rights. (3): Government Integrity. (4): Judicial Effectiveness. (5): Tax Burden. (6): Government Spending. (7): Fiscal Health. (8): Business Freedom. (9): Labor Freedom. (10): Monetary Freedom. (11): Trade Freedom. (12): Investment Freedom. (13): Financial Freedom. Source: www.heritage.org

3.2. Economic governance and capital restrictions

Our main explanatory variables consist of economic governance and capital restrictions. Given their notoriety in State governance studies, we use the Heritage Foundation's economic freedom indexes as a proxy for economic governance.⁹ In this manner, we regress BTC volumes on the thirteen freedom indexes, including the overall freedom index and the specific twelve freedom indexes. The Heritage indexes are scored from 0 to 100; the higher the score, the greater the freedom on four weights: Rule of Law, government size, regulatory efficiency, and open markets. Table 2 reports the average scores by country.

We resorted to the International Monetary Fund's capital restriction indexes concerning capital restrictions. We use a dataset, available at <http://www.columbia.edu/~mu2166/fkrsu/>, containing 57 different capital restrictions indexes by country by year for 1995–2019. These

indexes are presented in Fernández et al. (2016). For a given country, a restriction index is reported for each asset category (i.e., equity, bond, money market instructions, collective investments, derivatives, and real estate) and each type of capital movement ("Purchase locally by non-residents," "Sale or issue locally by nonresidents," "Purchase abroad by residents," and "Sale or issue abroad by residents"). For each country, we compute the average index across asset categories for each type of capital movement. Our average indexes by country are reported in Table 3.

3.3. Model and control variables

We estimate the following panel regression:

⁹ The datasets are openly available here: <https://www.heritage.org/index/explorer>.

Table 3
IMF restriction indexes by countries.

Countries	Restriction 1	Restriction 2
	Purchase Abroad by Residents	Sale or Issue Abroad by Residents
Argentina	0.571	0.571
Australia	0.000	0.000
Brazil	0.833	0.400
Canada	0.000	0.000
Chile	0.833	0.000
China	1.000	1.000
Colombia	0.833	0.800
Croatia	N/A	N/A
Czech Republic	1.000	0.200
Denmark	0.000	0.000
Dominican Republic	0.000	0.433
Egypt	0.833	0.200
Hong Kong	0.000	0.000
Hungary	0.167	0.000
India	1.000	0.800
Indonesia	0.667	0.800
Iran	0.000	0.000
Japan	0.000	0.000
Kazakhstan	0.556	0.500
Kenya	0.024	0.771
Malaysia	1.000	1.000
Mexico	1.000	0.200
Morocco	1.000	1.000
New Zealand	0.000	0.000
Nigeria	0.405	0.000
Norway	0.167	0.000
Pakistan	1.000	1.000
Peru	0.000	0.000
Philippines	0.976	1.000
Poland	1.000	0.800
Romania	0.000	0.000
Russia	0.095	0.800
Saudi Arabia	0.722	0.800
Singapore	0.000	0.000
South Africa	1.000	1.000
South Korea	0.167	0.114
Sweden	0.571	0.000
Switzerland	1.000	0.143
Tanzania	1.000	1.000
Thailand	1.000	0.800
Turkey	1.000	0.600
US	0.000	0.000
Ukraine	1.000	1.000
United Arab Emirates	0.000	0.000
United Kingdom	0.143	0.000
Venezuela	0.119	1.000
Vietnam	0.833	1.000

This table reports average indices by country for two categories of capital control (i.e., restrictions on capital movements). Large (low) values mean important (limited) capital control. The indices are presented in Fernández et al. (2016). The reported averages are calculated with the indices of six asset categories: equity, bond, money market instruments, collective investments, derivatives, and real estate. Data source: IMF.

$$\begin{aligned}
 Y1_{ij} = & \beta_0 + \beta_1 \times \text{ECONOMIC.GOVERNANCE.SCORE}_{ij} + \beta_2 \\
 & \times \text{CORRUPTION.INDEX}_{ij} + \beta_3 \times \text{RESTRICTION.PURCHASE}_{ij} \\
 & + \beta_4 \times \text{RESTRICTION.SALE}_{ij} + \beta_5 \times \text{GDP.PER.CAPITA}_{ij} + \beta_6 \\
 & \times \text{GDP.GROWTH}_{ij} + \beta_7 \times \text{INFLATION}_{ij} + \beta_8 \times \text{MONEY.GROWTH}_{ij} \\
 & + \beta_9 \times \text{UNEMPLOYMENT}_{ij} + \beta_{10} \times \text{WORLD.UNCERTAINTY.INDEX}_{ij} \\
 & + \beta_{11} \times \text{COUNTRY.RISK.PREMIUM}_{ij} + \beta_{12} \times \text{EXCHANGE.RATE}_{ij} \\
 & + \beta_{13} \times \text{BTC.RETURN.LAG}_{ij} + \varepsilon_{ij}
 \end{aligned}$$

Our model is estimated with ordinary least squares. ε_{ij} is the error term for country i and year j . Standard errors are corrected for heteroscedasticity following White (1980). Our two measures of capital control restrictions are somewhat stable over time for several countries, and even fully time-constant for a few of them. They exhibit low or no

within-variation, statistically speaking. Consequently, we use a simple OLS model with no fixed effects but with a correction for the heteroscedasticity in the regressions. Estimates with fixed and random effects on countries and years are exhibited in the Appendix, without capital control restrictions.

Under the hypothesis that BTC is censorship resistant, we expect countries with weak economic governance and severe corruption to be associated with large BTC trading volumes on the P2P platforms Localbitcoins and Paxful. Hence, we expect β_1 and β_2 to be negative. Likewise, countries with an important capital restriction program should be associated with large BTC trading volumes on these decentralized exchanges. We expect β_3 and β_4 to be positive. Given the strong correlation between the different economic governance scores of the Heritage Foundation, we ran several regressions by sequentially introducing the economic governance scores. We also test alternative models, replacing Y1 with Y2, Y3, and Y4. The results are discussed in the next section.

We use traditional macroeconomic aggregates as control variables, augmented with financial factors. We include GDP per capita, GDP growth, inflation, broad money growth (i.e., M2 stock), unemployment rate, and exchange rate change. All those variables are sourced from the World Bank. In addition to the traditional macroeconomic variables, we included two macroeconomic uncertainty proxies: the World Uncertainty Index (WUI), created by Ahir et al. (2022), and the country risk premium. The WUI is an index that tracks uncertainty in the world using text mining.¹⁰ The country risk premium per country is from Aswath Damodaran's website (<https://pages.stern.nyu.edu/~adamodar/>, Archives Data). It is defined as the country's default spread, based on agency ratings, times a multiplier that reflects the relative equity market volatility over the bond market's volatility. As for the exchange rate, we use end-of-year country currency against the US dollar. The last variable of the model is the annual bitcoin return lagged by one year. Descriptive statistics for these variables are presented in Table 4.

4. Empirical results

Given our four dependent variables, we run four different regressions. First, we regress economic governance, capital restrictions, and macroeconomic control variables on the sum of weekly BTC volume in the Localbitcoins DEX, as seen in Table 5. Of all economic governance indexes, only the overall economic and monetary freedom indexes are significant at the 1 % level. The tax burden index is significant at the 5 % level, while the judicial effectiveness, trade, investment, and financial freedom indexes are significant at the 10 % level. All significant economic freedom indexes negatively correlate with BTC volumes in Localbitcoins, except for judicial effectiveness, which correlates positively. When controlling for the economic freedom indexes, the average restrictions on assets purchased abroad by residents and the average restrictions on assets sold and issued abroad by residents are significant most of the time at the 1 % level and sometimes at the 10 % level (see Table 5).

The corruption perception index is negatively correlated with BTC volume when controlling for some governance proxies—judicial effectiveness, government spending, fiscal health, labor freedom at the 10 % level, and tax burden at the 1 % level. Indeed, less corrupt countries (highly scoring indexes) would result in less incentive for users to resort to a BTC DEX.

GDP per capita is the only macroeconomic control variable statistically significant when controlling for the economic governance proxies but at varying significance levels. It correlates positively with BTC volume, consistent with greater GDP per capita underlying market potential for trading assets. Other significant macroeconomic variables are broad money growth (when controlling for monetary freedom), unemployment rate (when controlling for investment freedom), and country

¹⁰ For more details, see <https://worlduncertaintyindex.com/data/>.

Table 4
Summary statistics on macroeconomic data.

Countries	Real GDP per capita	GDP growth	Inflation	Money Growth	Unemp-loyment rate	Country risk premium
Argentina	9.507	−0.183	N/A	33.623	8.202	9.045
Australia	10.956	2.480	1.885	6.769	5.650	0.000
Brazil	9.079	0.145	5.878	9.347	10.107	3.612
Canada	10.692	2.079	1.602	N/A	6.531	0.000
Chile	9.515	2.161	3.117	7.348	6.800	0.854
China	9.048	6.961	2.078	10.466	4.514	0.870
Colombia	8.727	3.112	4.071	8.897	8.936	2.650
Croatia	9.437	2.164	0.545	3.533	12.869	3.869
Czech Republic	9.817	3.077	1.460	7.034	4.171	0.941
Denmark	10.906	2.152	0.682	2.844	6.080	0.000
Dominican Republic	8.911	6.223	2.351	9.858	6.610	5.486
Egypt	8.206	4.125	13.826	19.972	11.587	9.129
Hong Kong	10.672	2.199	2.992	7.688	3.183	0.592
Hungary	9.488	3.730	1.482	7.283	5.874	3.188
India	7.435	6.719	5.362	10.614	5.382	2.843
Indonesia	8.150	5.105	4.676	9.256	4.157	2.892
Iran	8.598	1.923	17.051	29.142	11.605	N/A
Japan	10.471	0.948	0.815	3.018	3.100	0.934
Kazakhstan	9.286	3.200	7.770	11.242	4.917	2.867
Kenya	7.308	4.657	6.201	12.416	3.416	6.461
Malaysia	9.239	5.048	2.123	4.936	3.221	1.633
Mexico	9.176	2.036	3.992	9.350	4.010	1.718
Morocco	7.971	3.259	1.197	4.728	9.354	3.398
New Zealand	10.574	3.308	1.196	7.110	5.001	0.000
Nigeria	7.857	2.708	11.605	10.334	6.429	6.240
Norway	11.222	1.461	2.384	4.859	3.933	0.000
Pakistan	7.256	4.669	5.538	11.463	3.453	8.941
Peru	8.762	3.053	2.854	7.582	3.463	1.605
Philippines	8.061	6.570	2.666	13.508	2.857	2.650
Poland	9.482	3.830	0.803	7.876	6.429	1.154
Romania	9.169	4.431	1.815	9.639	5.663	2.993
Russia	9.165	1.078	6.884	10.641	5.187	3.035
Saudi Arabia	9.914	1.923	0.840	3.857	5.760	0.915
Singapore	10.959	3.530	0.558	4.181	3.773	0.000
South Africa	8.736	1.235	5.264	6.819	26.223	2.672
South Korea	10.292	2.919	1.151	6.831	3.464	0.720
Sweden	10.852	2.416	0.892	6.347	7.190	0.000
Switzerland	11.359	1.955	0.003	3.008	4.743	0.000
Tanzania	6.920	6.212	4.608	8.883	2.181	5.344
Thailand	8.711	2.970	0.829	4.849	0.634	2.173
Turkey	9.327	4.886	10.635	18.322	10.724	3.704
US	10.960	2.281	1.550	4.842	5.089	0.000
Ukraine	7.741	−1.188	15.389	9.363	8.774	12.047
United Arab Emirates	10.561	3.498	1.748	7.720	2.058	0.677
United Kingdom	10.734	2.175	1.673	3.213	5.116	0.592
Venezuela	N/A	−1.276	119.874	58.843	5.853	14.823
Vietnam	7.941	6.898	2.873	15.517	1.672	5.659

This table reports the average values of macroeconomics indicators by country for the period 2013–2019. The numbers of columns 3 to 7 are in percentage. Data Source: World Bank.

risk premium (when controlling for fiscal health and monetary freedom). Except for country risk premium when controlling for monetary freedom, all the others are significant only at the 10 % level. Country risk premium is significant at the 1 % level when controlling for monetary freedom, which is coherent with the fact that countries with more monetary freedom are also risk-safer.

Since Paxful data only starts in 2015, we run a second model like the first. However, we consolidate BTC transaction volume data from Localbitcoins and Paxful with results in Table 6. Including Paxful data makes specific governance proxies insignificant, while the other previously significant proxies remain significant and retain their correlation sign. The now insignificant governance proxies are judicial effectiveness and investment freedom. Concerning the corruption perceptions index, it becomes positively significant at the 10 % level in regressions 8 and 11 of the table. Meanwhile, the results do not change the significance levels and signs of capital restriction variables. Similarly, GDP per capita, broad money growth, and country risk premium face little change when Paxful data is included in the regression. At the same time, the

unemployment rate has now become insignificant.

For robustness, we run two additional regressions, using the sum of weekly transactions divided by the average weekly outstanding BTC, thus leveling the data relative to the outstanding BTC stock in circulation in both DEXs and CEXs (and outside exchanges). The first additional regression (see Table 7) includes only Localbitcoins transaction volumes. The significance and sign orientation of governance proxies is the same, except for investment freedom, which is statistically insignificant. Regarding significance and signs, the results for capital restrictions are much the same as before. Still, residents' average number of assets sold and issued abroad is not insignificant. The corruption perceptions index sees its significance reduced to only three regressions (judicial effectiveness, tax burden, and fiscal health), with no change in signs. Concerning macroeconomic variables, the unemployment rate is positively significant at the 10 % level when controlling for investment and financial freedoms. Country risk premium remains negatively significant in regressions 7 and 10 of the table.

Finally, the second additional model includes the Paxful database

Table 5
Annual Localbitcoins data regression.

Dependent variable: Y1 (Total Localbitcoins volume by country-year)														
Overall Score	−1.899*** (0.598)													
Property Rights		−0.261 (0.326)												
Government Integrity			−1.412 (0.968)											
Judicial Effectiveness				0.645** (0.256)										
Tax Burden					−0.936*** (0.345)									
Government Spending						−0.242* (0.142)								
Fiscal Health							−0.165 (0.116)							
Business Freedom								−0.110 (0.302)						
Labor Freedom									0.613 (0.512)					
Monetary Freedom										−3.798*** (1.006)				
Trade Freedom											−0.687** (0.342)			
Investment Freedom												−0.541** (0.261)		
Financial Freedom													−0.681** (0.278)	
Corruption Index (CPI)	−0.115 (0.252)	−0.337 (0.287)	0.885 (0.981)	−0.955*** (0.367)	−0.934*** (0.337)	−0.637** (0.290)	−0.544** (0.269)	−0.507* (0.281)	−0.564* (0.299)	0.158 (0.246)	−0.481* (0.270)	−0.323 (0.238)	−0.234 (0.240)	
Average Restrictions PABR	−35.276*** (9.612)	−34.621*** (9.608)	−36.799*** (10.764)	−17.739** (8.865)	−37.058*** (10.345)	−35.770*** (9.890)	−18.433** (9.251)	−35.164*** (10.007)	−30.110*** (8.846)	−35.017*** (8.820)	−33.733*** (9.683)	−32.112*** (8.913)	−31.533*** (9.038)	
Average Restrictions SIAR	23.725*** (8.124)	28.372*** (8.863)	32.669*** (10.305)	28.439** (12.290)	29.239*** (9.178)	28.661*** (9.184)	32.309** (13.288)	29.857*** (9.759)	30.220*** (9.566)	15.868** (7.215)	28.427*** (8.972)	17.257** (8.273)	22.500*** (7.813)	
GDP per Capita	21.070*** (7.845)	18.903** (7.637)	20.403** (8.405)	7.779 (4.986)	21.836*** (8.301)	17.078** (7.638)	11.946** (5.707)	19.666** (8.389)	16.758** (6.665)	9.786* (5.387)	21.112*** (8.017)	17.940** (7.086)	17.091** (7.023)	
GDP Growth	−1.016 (1.207)	−1.044 (1.249)	−1.087 (1.216)	−3.395 (2.320)	−0.461 (1.240)	−0.745 (1.208)	−2.716 (2.333)	−0.999 (1.248)	−0.995 (1.230)	−1.376 (1.269)	−1.029 (1.215)	−0.630 (1.195)	−1.444 (1.236)	
Inflation	1.368* (0.772)	1.552* (0.859)	1.704* (0.875)	1.333 (1.474)	1.808** (0.882)	1.621* (0.859)	2.219 (1.474)	1.483* (0.828)	1.172 (0.803)	−0.013 (0.744)	1.374* (0.770)	0.865 (0.743)	1.165 (0.763)	
Broad Money Growth	−1.010 (0.648)	−1.100 (0.698)	−0.994 (0.652)	0.667 (0.930)	−1.039 (0.685)	−0.982 (0.659)	0.515 (0.910)	−1.023 (0.655)	−0.956 (0.643)	−1.312** (0.659)	−1.061 (0.646)	−0.938 (0.624)	−0.945 (0.627)	
Unemployment Rate	0.450 (0.469)	0.854* (0.486)	0.711 (0.447)	−0.781 (0.511)	0.190 (0.448)	0.641 (0.478)	−0.821* (0.478)	0.797 (0.491)	1.040* (0.620)	0.327 (0.391)	0.720 (0.482)	1.022* (0.544)	0.832 (0.506)	
World Uncertainty Index	−0.847* (0.434)	−0.794* (0.425)	−0.799* (0.413)	−0.823* (0.486)	−0.955** (0.455)	−0.900** (0.443)	−0.747 (0.479)	−0.902** (0.446)	−0.856** (0.425)	−0.885** (0.425)	−0.886** (0.443)	−0.800* (0.422)	−0.938** (0.449)	
Country Risk Premium	−2.443** (1.156)	−1.380 (1.242)	−1.464 (1.065)	−1.935 (1.565)	−0.772 (1.191)	−1.691 (1.240)	−3.443** (1.738)	−0.929 (1.147)	−0.298 (1.465)	−3.759*** (1.171)	−0.900 (1.085)	−1.049 (1.085)	−2.096* (1.171)	
Exchange Rate Change	−0.536 (0.363)	−0.563 (0.375)	−0.565 (0.375)	−0.486 (0.437)	−0.620 (0.383)	−0.576 (0.366)	−0.605 (0.453)	−0.533 (0.365)	−0.498 (0.362)	0.049 (0.294)	−0.496 (0.361)	−0.433 (0.343)	−0.500 (0.357)	
BTC return	−1.619* (0.890)	−1.683* (0.889)	−1.767* (0.919)	−0.772 (0.552)	−1.933** (0.958)	−1.793* (0.929)	−0.547 (0.524)	−1.797* (0.929)	−1.755* (0.901)	−1.236 (0.783)	−1.748* (0.926)	−1.633* (0.879)	−1.739* (0.910)	
Intercept	4.866 (44.625)	−83.692* (48.998)	−104.129* (57.770)	−11.645 (39.136)	−19.081 (45.461)	−47.445 (56.177)	−27.228 (42.383)	−88.448* (50.286)	−113.316* (64.508)	275.573*** (74.733)	−55.726 (49.023)	−55.987 (40.274)	−38.970 (43.659)	
Number of observations	238	238	238	118	238	238	118	238	238	238	238	238	238	
Adjusted R-square	5.1	3.7	4.4	8.3	5.3	3.9	6.6	3.6	4.6	10.4	3.8	4.5	4.6	

This table reports regression results where the dependent variable Y1 is regressed on Heritage indexes and a set of control variables. Y1 is the annual volume of BTC traded on Localbitcoins, scaled by one thousand. See Subsection 3.1 for more details. Standard errors, in parentheses, follow coefficient estimates. Standard errors are corrected for heteroscedasticity following White (1980). ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

Table 6
Annual Localbitcoins and Paxful data regression.

Dependent Variable: Y2 (Total Localbitcoins and Paxful volume by country-year)														
Overall Score	−2.058*** (0.645)													
Property Rights		−0.333 (0.340)												
Government Integrity			−1.332 (1.002)											
Judicial Effectiveness				0.644** (0.298)										
Tax Burden					−1.051*** (0.383)									
Government Spending						−0.237 (0.153)								
Fiscal Health							−0.356* (0.188)							
Business Freedom								−0.249 (0.341)						
Labor Freedom									0.761 (0.534)					
Monetary Freedom										−4.045*** (1.048)				
Trade Freedom											−0.734** (0.364)			
Investment Freedom												−0.503* (0.277)		
Financial Freedom													−0.633** (0.298)	
Corruption Index	−0.154 (0.271)	−0.358 (0.305)	0.730 (1.009)	−1.017** (0.396)	−1.058*** (0.359)	−0.709** (0.303)	−0.659** (0.298)	−0.549* (0.290)	−0.646** (0.317)	0.128 (0.259)	−0.552* (0.284)	−0.412 (0.259)	−0.330 (0.267)	
Average Restrictions PABR	−38.359*** (10.115)	−37.601*** (10.071)	−39.739*** (11.261)	−22.954** (9.871)	−40.374*** (10.916)	−38.803*** (10.359)	−23.372** (10.352)	−38.583*** (10.666)	−32.009*** (9.109)	−38.076*** (9.304)	−36.705*** (10.177)	−35.355*** (9.460)	−34.821*** (9.600)	
Average Restrictions SIAR	25.234*** (8.360)	30.102*** (9.061)	34.444*** (10.569)	31.993** (12.471)	31.208*** (9.404)	30.643*** (9.426)	38.826*** (14.283)	32.600*** (10.242)	32.425*** (9.868)	16.970** (7.481)	30.341*** (9.177)	20.068** (8.854)	24.952*** (8.204)	
GDP per Capita	25.056*** (8.571)	22.705*** (8.361)	24.124*** (9.116)	12.397* (6.464)	25.998*** (9.090)	20.918** (8.425)	18.863** (7.964)	24.422*** (9.432)	20.043*** (7.299)	12.998** (5.983)	25.068*** (8.757)	21.814*** (7.872)	21.027*** (7.842)	
GDP Growth	−0.991 (1.287)	−1.046 (1.329)	−1.043 (1.295)	−2.965 (2.396)	−0.372 (1.313)	−0.714 (1.286)	−2.358 (2.478)	−1.074 (1.346)	−0.980 (1.317)	−1.372 (1.376)	−1.003 (1.294)	−0.616 (1.263)	−1.372 (1.316)	
Inflation	1.653* (0.844)	1.862** (0.936)	1.989** (0.948)	2.077 (1.696)	2.142** (0.967)	1.916** (0.930)	3.210* (1.777)	1.755* (0.900)	1.389 (0.859)	0.185 (0.822)	1.662** (0.846)	1.206 (0.830)	1.486* (0.842)	
Broad Money Growth	−1.005 (0.672)	−1.118 (0.718)	−0.992 (0.674)	0.632 (1.048)	−1.037 (0.714)	−0.979 (0.682)	0.551 (1.035)	−1.022 (0.679)	−0.936 (0.671)	−1.327* (0.684)	−1.061 (0.667)	−0.940 (0.646)	−0.947 (0.650)	
Unemployment Rate	0.390 (0.489)	0.835* (0.499)	0.687 (0.463)	−0.819 (0.531)	0.083 (0.475)	0.614 (0.494)	−1.057* (0.551)	0.745 (0.510)	1.065* (0.639)	0.265 (0.410)	0.683 (0.498)	0.978* (0.557)	0.801 (0.520)	
World Uncertainty Index	−0.703 (0.486)	−0.625 (0.480)	−0.665 (0.468)	−0.697 (0.626)	−0.822 (0.507)	−0.759 (0.495)	−0.608 (0.620)	−0.766 (0.498)	−0.706 (0.477)	−0.744 (0.473)	−0.745 (0.495)	−0.667 (0.475)	−0.795 (0.504)	
Country Risk Premium	−2.484** (1.250)	−1.413 (1.319)	−1.352 (1.153)	−2.417 (1.934)	−0.666 (1.279)	−1.594 (1.336)	−4.654** (2.254)	−0.812 (1.241)	−0.056 (1.545)	−3.858*** (1.291)	−0.813 (1.164)	−0.959 (1.155)	−1.932 (1.263)	
Exchange Rate Change	−0.653* (0.393)	−0.688* (0.406)	−0.681* (0.405)	−0.669 (0.465)	−0.747* (0.417)	−0.693* (0.395)	−0.887* (0.523)	−0.648 (0.395)	−0.606 (0.390)	−0.030 (0.321)	−0.611 (0.391)	−0.558 (0.374)	−0.620 (0.388)	
BTC return	−1.594* (0.957)	−1.642* (0.960)	−1.759* (0.988)	−0.754 (0.730)	−1.940* (1.026)	−1.784* (0.999)	−0.459 (0.685)	−1.783* (0.998)	−1.734* (0.968)	−1.190 (0.850)	−1.735* (0.996)	−1.636* (0.951)	−1.734* (0.982)	
Intercept	−20.520 (50.763)	−115.577** (55.531)	−136.432** (64.231)	−53.266 (54.139)	−43.803 (51.770)	−81.415 (63.952)	−73.656 (59.484)	−121.591** (57.163)	−152.506** (71.671)	266.038*** (77.307)	−86.687 (55.509)	−91.459* (49.289)	−75.691 (53.183)	
Number of observations	238	238	238	118	238	238	118	238	238	238	238	238	238	
Adjusted R-square	5.6	4.2	4.6	4.4	5.9	4.3	7.0	4.1	5.4	10.6	4.3	4.7	4.8	

This table reports regression results where the dependent variable Y2 is regressed on Heritage indexes and a set of control variables. Y2 is the annual volume of BTC traded on Localbitcoins and Paxful, scaled by one thousand. See [Subsection 3.1](#) for more details. Standard errors, in parentheses, follow coefficient estimates. Standard errors are corrected for heteroscedasticity following [White \(1980\)](#). ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

Table 7
Annual scaled Localbitcoins data regression.

Dependent Variable: Y3 (Localbitcoins volume scaled by outstanding BTC)														
Overall Score	−0.123*** (0.039)													
Property Rights		−0.017 (0.021)												
Government Integrity			−0.103 (0.066)											
Judicial Effectiveness				0.038** (0.015)										
Tax Burden					−0.063*** (0.023)									
Government Spending						−0.016* (0.010)								
Fiscal Health							−0.010 (0.007)							
Business Freedom								−0.006 (0.019)						
Labor Freedom									0.043 (0.035)					
Monetary Freedom										−0.253*** (0.069)				
Trade Freedom											−0.043* (0.022)			
Investment Freedom												−0.034** (0.017)		
Financial Freedom													−0.042** (0.018)	
Corruption Index	−0.005 (0.016)	−0.019 (0.018)	0.071 (0.066)	−0.056** (0.022)	−0.059*** (0.022)	−0.039** (0.019)	−0.032** (0.016)	−0.031* (0.018)	−0.035* (0.019)	0.014 (0.016)	−0.029* (0.017)	−0.019 (0.015)	−0.014 (0.016)	
Average Restrictions PABR	−2.333*** (0.637)	−2.290*** (0.635)	−2.447*** (0.718)	−1.052** (0.531)	−2.452*** (0.686)	−2.366*** (0.655)	−1.093** (0.553)	−2.321*** (0.660)	−1.970*** (0.571)	−2.316*** (0.586)	−2.235*** (0.641)	−2.131*** (0.589)	−2.102*** (0.600)	
Average Restrictions SIAR	1.509*** (0.516)	1.810*** (0.561)	2.118*** (0.669)	1.690** (0.735)	1.868*** (0.583)	1.829*** (0.584)	1.920** (0.795)	1.899*** (0.620)	1.937*** (0.612)	0.977** (0.465)	1.817*** (0.570)	1.106** (0.534)	1.455*** (0.501)	
GDP per Capita	1.360*** (0.522)	1.220** (0.508)	1.329** (0.564)	0.458 (0.298)	1.416** (0.553)	1.098** (0.509)	0.705** (0.341)	1.259** (0.554)	1.068** (0.436)	0.613* (0.358)	1.358** (0.532)	1.158** (0.472)	1.109** (0.470)	
GDP Growth	−0.061 (0.079)	−0.063 (0.081)	−0.067 (0.079)	−0.202 (0.140)	−0.024 (0.081)	−0.043 (0.079)	−0.162 (0.140)	−0.059 (0.082)	−0.060 (0.080)	−0.085 (0.083)	−0.062 (0.079)	−0.036 (0.078)	−0.087 (0.081)	
Inflation	0.087* (0.050)	0.099* (0.056)	0.110* (0.057)	0.079 (0.088)	0.116** (0.058)	0.103* (0.056)	0.131 (0.088)	0.094* (0.054)	0.072 (0.052)	−0.006 (0.049)	0.087* (0.050)	0.055 (0.049)	0.075 (0.050)	
Broad Money Growth	−0.069 (0.044)	−0.075 (0.047)	−0.068 (0.044)	0.041 (0.055)	−0.071 (0.046)	−0.067 (0.044)	0.032 (0.054)	−0.070 (0.044)	−0.065 (0.043)	−0.089** (0.045)	−0.072* (0.043)	−0.065 (0.042)	−0.065 (0.042)	
Unemployment Rate	0.034 (0.032)	0.060* (0.033)	0.050 (0.030)	−0.047 (0.031)	0.016 (0.030)	0.046 (0.032)	−0.049* (0.029)	0.056* (0.033)	0.073* (0.043)	0.025 (0.026)	0.051 (0.033)	0.070* (0.037)	0.058* (0.034)	
World Uncertainty Index	−0.064** (0.030)	−0.060** (0.029)	−0.060** (0.028)	−0.052* (0.029)	−0.071** (0.031)	−0.067** (0.030)	−0.048* (0.029)	−0.067** (0.031)	−0.064** (0.029)	−0.066** (0.029)	−0.066** (0.030)	−0.061** (0.029)	−0.069** (0.031)	
Country Risk Premium	−0.156** (0.077)	−0.088 (0.083)	−0.097 (0.072)	−0.114 (0.093)	−0.047 (0.080)	−0.109 (0.083)	−0.203** (0.103)	−0.058 (0.077)	−0.013 (0.100)	−0.246*** (0.078)	−0.056 (0.074)	−0.066 (0.073)	−0.130* (0.078)	
Exchange Rate Change	−0.033 (0.024)	−0.035 (0.025)	−0.035 (0.025)	−0.029 (0.026)	−0.039 (0.025)	−0.036 (0.024)	−0.036 (0.027)	−0.033 (0.024)	−0.030 (0.024)	0.006 (0.019)	−0.031 (0.024)	−0.027 (0.023)	−0.031 (0.024)	
BTC return	−0.112* (0.060)	−0.116* (0.060)	−0.122** (0.062)	−0.049 (0.033)	−0.133** (0.065)	−0.124** (0.063)	−0.035 (0.032)	−0.124** (0.063)	−0.121** (0.061)	−0.087 (0.053)	−0.121* (0.063)	−0.114* (0.059)	−0.120* (0.062)	
Intercept	0.549 (2.957)	−5.196 (3.230)	−6.659* (3.871)	−0.571 (2.335)	−0.878 (2.995)	−2.785 (3.748)	−1.494 (2.527)	−5.514* (3.322)	−7.270* (4.347)	18.723*** (5.087)	−3.463 (3.239)	−3.449 (2.663)	−2.480 (2.905)	
Number of observations	238	238	238	118	238	238	118	238	238	238	238	238	238	
Adjusted R-square	4.9	3.5	4.4	8.1	5.1	3.8	6.6	3.5	4.5	10.0	3.7	4.3	4.2	

This table reports regression results where the dependent variable Y3 is regressed on Heritage indexes and a set of control variables. Y3 is the annual volume of BTC traded on Localbitcoins, scaled by the annual average of outstanding BTC. See [Subsection 3.1](#) for more details. Standard errors, in parentheses, follow coefficient estimates. Standard errors are corrected for heteroscedasticity following [White \(1980\)](#). ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

Table 8
Annual scaled Localbitcoins and Paxful data regression.

Dependent Variable: Y4 (Localbitcoins and Paxful volume scaled by outstanding BTC)														
Overall Score	−0.133*** (0.042)													
Property Rights		−0.022 (0.022)												
Government Integrity			−0.099 (0.068)											
Judicial Effectiveness				0.038** (0.018)										
Tax Burden					−0.070*** (0.026)									
Government Spending						−0.016 (0.010)								
Fiscal Health							−0.021* (0.011)							
Business Freedom								−0.014 (0.022)						
Labor Freedom									0.052 (0.037)					
Monetary Freedom										−0.268*** (0.071)				
Trade Freedom											−0.046* (0.024)			
Investment Freedom												−0.032* (0.018)		
Financial Freedom													−0.039** (0.019)	
Corruption Index	−0.007 (0.018)	−0.020 (0.019)	0.063 (0.067)	−0.059** (0.024)	−0.067*** (0.023)	−0.044** (0.019)	−0.038** (0.018)	−0.033* (0.019)	−0.039* (0.020)	0.012 (0.017)	−0.033* (0.018)	−0.024 (0.017)	−0.020 (0.017)	
Average Restrictions PABR	−2.518*** (0.666)	−2.469*** (0.663)	−2.625*** (0.748)	−1.358** (0.591)	−2.652*** (0.720)	−2.548*** (0.683)	−1.382** (0.618)	−2.527*** (0.699)	−2.084*** (0.587)	−2.500*** (0.614)	−2.414*** (0.670)	−2.325*** (0.621)	−2.298*** (0.632)	
Average Restrictions SIAR	1.598*** (0.530)	1.913*** (0.573)	2.226*** (0.686)	1.898** (0.746)	1.985*** (0.597)	1.947*** (0.599)	2.300*** (0.853)	2.063*** (0.649)	2.068*** (0.631)	1.042** (0.482)	1.931*** (0.583)	1.269** (0.566)	1.598*** (0.524)	
GDP per Capita	1.596*** (0.563)	1.444*** (0.549)	1.550** (0.605)	0.725* (0.382)	1.662*** (0.598)	1.325** (0.554)	1.105** (0.470)	1.540** (0.613)	1.261*** (0.472)	0.802** (0.391)	1.592*** (0.574)	1.387*** (0.517)	1.340*** (0.516)	
GDP Growth	−0.060 (0.084)	−0.064 (0.086)	−0.065 (0.084)	−0.177 (0.144)	−0.019 (0.086)	−0.042 (0.084)	−0.142 (0.149)	−0.064 (0.088)	−0.060 (0.086)	−0.085 (0.089)	−0.060 (0.084)	−0.036 (0.082)	−0.083 (0.085)	
Inflation	0.104* (0.055)	0.117* (0.061)	0.127** (0.062)	0.122 (0.101)	0.136** (0.063)	0.121** (0.060)	0.189* (0.105)	0.111* (0.058)	0.085 (0.056)	0.006 (0.054)	0.105* (0.055)	0.075 (0.053)	0.094* (0.055)	
Broad Money Growth	−0.069 (0.045)	−0.077 (0.048)	−0.068 (0.045)	0.038 (0.062)	−0.071 (0.048)	−0.067 (0.046)	0.033 (0.061)	−0.070 (0.045)	−0.064 (0.045)	−0.090** (0.046)	−0.073 (0.045)	−0.065 (0.043)	−0.066 (0.044)	
Unemployment Rate	0.030 (0.033)	0.059* (0.034)	0.048 (0.031)	−0.049 (0.032)	0.009 (0.031)	0.044 (0.033)	−0.063* (0.033)	0.053 (0.035)	0.075* (0.044)	0.021 (0.027)	0.049 (0.034)	0.068* (0.038)	0.057 (0.035)	
World Uncertainty Index	−0.056* (0.032)	−0.051 (0.032)	−0.053* (0.031)	−0.046 (0.038)	−0.064* (0.034)	−0.060* (0.033)	−0.041 (0.037)	−0.060* (0.033)	−0.056* (0.032)	−0.059* (0.032)	−0.059* (0.033)	−0.054* (0.032)	−0.062* (0.034)	
Country Risk Premium	−0.159* (0.083)	−0.090 (0.087)	−0.091 (0.077)	−0.143 (0.114)	−0.041 (0.085)	−0.103 (0.089)	−0.274** (0.133)	−0.052 (0.083)	0.001 (0.105)	−0.253*** (0.085)	−0.051 (0.078)	−0.061 (0.077)	−0.121 (0.083)	
Exchange Rate Change	−0.040 (0.026)	−0.042 (0.027)	−0.042 (0.027)	−0.040 (0.028)	−0.046* (0.027)	−0.043* (0.026)	−0.052* (0.031)	−0.040 (0.026)	−0.037 (0.026)	0.001 (0.021)	−0.038 (0.026)	−0.034 (0.024)	−0.038 (0.025)	
BTC return	−0.112* (0.064)	−0.115* (0.064)	−0.122* (0.066)	−0.048 (0.044)	−0.134* (0.069)	−0.124* (0.067)	−0.031 (0.041)	−0.124* (0.067)	−0.120* (0.065)	−0.084 (0.057)	−0.121* (0.067)	−0.114* (0.063)	−0.121* (0.066)	
Intercept	−0.909 (3.279)	−7.051** (3.581)	−8.548** (4.225)	−2.943 (3.159)	−2.294 (3.331)	−4.767 (4.172)	−4.141 (3.474)	−7.443** (3.692)	−9.564** (4.732)	18.218*** (5.227)	−5.266 (3.588)	−5.507* (3.135)	−4.604 (3.398)	
Number of observations	238	238	238	118	238	238	118	238	238	238	238	238	238	
Adjusted R-square	5.3	4.0	4.6	4.3	5.7	4.2	6.8	3.9	5.3	10.3	4.1	4.5	4.5	

This table reports regression results where the dependent variable Y4 is regressed on Heritage indexes and a set of control variables. Y4 is the annual volume of BTC traded on Localbitcoins and Paxful, scaled by the annual average of outstanding BTC. See [Subsection 3.1](#) for more details. Standard errors, in parentheses, follow coefficient estimates. Standard errors are corrected for heteroscedasticity following [White \(1980\)](#). ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

and corroborates much of the results above (see Table 8). In this manner, our most significant results for governance proxies, capital restrictions, corruption, GDP per capita, and country risk premium do not change signs. They remain strongly significant in all models tested.¹¹

5. Discussion

Our results indicate that BTC users resort to DEXs when concerned with threats to overall economic freedoms, monetary governance, and tax burden. This is expected since resorting to unregulated platforms would be incentivized by heavy tax burdens and loose monetary governance. Indeed, one of BTC's promises is to escape the supposedly mismanaged fiat currency monetary system and government scrutiny of one's financial transaction, primarily observed in unregulated DEXs relative to more regulated CEXs.

The results of capital restrictions show two different movements in DEXs' transaction volumes. The average of purchased assets abroad by residents negatively correlates with BTC transaction volumes. One would think that unregulated BTC trades would correlate positively with greater restrictions. However, acquisitions of foreign assets mainly necessitate access to officially-sanctioned banking networks. The same remains true for BTC networks despite the promise of independence from fiat monetary systems. Although some DEX transactions can involve exotic exchange media such as collectibles and Amazon® coupons, these remain marginal. Most transactions in DEXs remain anchored to fiat currencies. As long as access to the fiat monetary system is necessary, it is easy for a restrictive State to control capital flows to foreign asset networks, among which BTC. Consequently, the tighter the restrictions on asset purchases abroad, the lower the BTC transaction volumes in DEXs. This result is consistent with our data, which shows the greatest transaction volumes in developed countries where capital restrictions are mostly weak.

Residents' restrictions on sales and issues of assets abroad vary directly with BTC transaction volumes in DEXs. This is consistent with the fact that restricting sales and issues of domestic assets only reduces their marketability while increasing that of international assets. If domestic investors cannot resort to foreign markets, only BTC remains a tradable asset despite purchase restrictions. Indeed, it might not be easy for residents to buy BTC abroad, but networks can form where BTC is tradable internally. Once BTC is acquired, it becomes tradable in DEXs, CEXs, and even physically (on hardware storage). Moreover, domestic controls can be bypassed once BTC is in hand via VPNs (Virtual Private Networks).

The more isolated results concerning country risk premium—primarily significant when controlling for monetary freedom—indicate a weakness of the domestic currency. As risk premium increases, BTC transaction volumes fall in DEXs because the domestic currency tends to depreciate under that circumstance. Risk premium would be mostly a problem concerning monetarily free countries, where trading on foreign exchange is largely unregulated.

Given the above significant results, one could expect similar results to the relationship between BTC trade volumes and variables such as property rights, government integrity, and world uncertainty and corruption indexes. However, the results are all insignificant. Property rights tend to be stable over time and may not vary enough within countries to have a measurable impact on BTC transaction volumes in DEXs. Moreover, BTC transactions on DEXs are often used to circumvent

formal institutional frameworks, meaning individuals engaging in these transactions may not directly rely on or be influenced by the strength of property rights.

Government integrity, which reflects transparency and the absence of corruption, may not have a direct or strong enough relationship with BTC trade volumes. Individuals might resort to DEXs due to other governance issues, such as economic instability or capital restrictions, rather than a specific perception of governmental corruption or integrity. This finding highlights that DEX usage is likely influenced by practical, immediate constraints (e.g., monetary or regulatory policies) rather than broader ethical or transparency concerns within governance.

While corruption can influence financial behavior, it may be that the corruption index lacks the granularity to capture the specific dynamics influencing DEX usage. For example, highly corrupt countries may also have informal networks that allow for alternative forms of capital mobility, reducing the need for DEX transactions.

Concerning the world uncertainty index, it captures global trends, which may not reflect localized uncertainty that more directly impacts BTC trading behaviors. Users on DEXs may be responding to specific national or regional uncertainties (e.g., capital controls, currency devaluations) rather than global uncertainty. Moreover, the effects of global uncertainty might vary across countries depending on their institutional resilience and economic structure.

Finally, another interesting finding is the insignificance of BTC return on BTC transaction volumes in DEXs. In fact, DEX users often engage in BTC trading for reasons unrelated to its speculative returns, such as avoiding capital controls or seeking censorship-resistant transactions. Furthermore, DEX users may differ in their motivations, with some focused on its utility as a medium of exchange or store of value rather than short-term profit from price movements.

In sum, our results show that BTC transactions in DEXs are resorted to as an alternative to weak monetary governance and a restrictive financial system.

6. Additional analyses

6.1. Selection bias

The coverage by Localbitcoins and Paxful of a country might be related to some specific (observable or not) factors. If this is the case, our sample of countries is non-randomly selected from the complete list of countries in the world, and our analysis of the determinants of BTC trading volume with our sample produces wrong estimates, leading to erroneous conclusions. We employ the Heckman two-step approach to correct the potential bias in our sample of countries. First, we run the following probit regression on panel data:

$$\text{Prob}\{INCLUDED_{it} = 1 \mid \bullet\} = N\left(\alpha_0 + \sum_z \alpha_z \times EXPLANATORY_VARIABLES_{z,ij} + e_{ij}\right).$$

INCLUDED takes the value 1 on year t if the country i is included in our sample, 0 otherwise. $N()$ denotes the normal cumulative distribution. EXPLANATORY_VARIABLES denotes all the variables used in our previous regressions. The results of the probit regressions are presented in Table 9. Then, we calculate the “lambda” statistics. This is the ratio of the probability density function to the cumulative distribution function of the normal distribution. The density and cumulative distribution are both measured at the linear combination of the parameters for each firm-year observation:

$$LAMBDA_{ij} = \frac{f(\hat{\theta}_{ij})}{N(\hat{\theta}_{ij})}$$

with

¹¹ In an additional set of regressions, available upon request, the sample is expanded to include data from 2020, resulting in an adjustment of the total country-year observations from 238 to 294. Due to the unavailability of capital control measures and country risk premia beyond 2019, these variables are excluded from the analysis for this extended period. The results remain consistent with the original findings in terms of the direction and magnitude of the coefficients.

Table 9
Heckman's first stage - probit regression.

Dependent Variable: Y1 (Total Localbitcoins volume by country-year)													
Overall Score	0.010 (0.016)												
Property Rights		0.004 (0.017)											
Government Integrity			0.013 (0.016)										
Judicial Effectiveness				0.020 (0.023)									
Tax Burden					0.010 (0.017)								
Government Spending						0.012 (0.016)							
Fiscal Health							0.033 (0.022)						
Business Freedom								0.012 (0.016)					
Labor Freedom									0.012 (0.016)				
Monetary Freedom										0.010 (0.016)			
Trade Freedom											0.011 (0.016)		
Investment Freedom												0.011 (0.016)	
Financial Freedom													0.010 (0.017)
Corruption Index	0.742*** (0.247)	0.730*** (0.247)	0.711*** (0.242)	0.673* (0.378)	0.623** (0.246)	0.665*** (0.244)	0.626* (0.375)	0.693*** (0.242)	0.819*** (0.250)	0.702*** (0.244)	0.701*** (0.244)	0.705*** (0.244)	0.600** (0.250)
Average Restrictions PABR	0.651** (0.265)	0.528** (0.267)	0.601** (0.262)	0.873** (0.406)	0.559** (0.262)	0.635** (0.262)	0.902** (0.399)	0.599** (0.266)	0.599** (0.264)	0.662** (0.266)	0.665** (0.268)	0.659** (0.270)	0.916*** (0.280)
Average Restrictions SIAR	−0.262*** (0.044)	−0.256*** (0.045)	−0.271*** (0.044)	−0.402*** (0.087)	−0.265*** (0.044)	−0.281*** (0.045)	−0.467*** (0.093)	−0.276*** (0.044)	−0.265*** (0.044)	−0.270*** (0.044)	−0.276*** (0.044)	−0.274*** (0.044)	−0.251*** (0.044)
GDP per Capita	−0.005 (0.011)	−0.012 (0.012)	−0.005 (0.011)	0.015 (0.018)	−0.006 (0.011)	−0.005 (0.011)	0.018 (0.018)	−0.005 (0.011)	−0.004 (0.011)	−0.005 (0.011)	−0.005 (0.011)	−0.005 (0.011)	−0.003 (0.011)
GDP Growth	−0.005 (0.008)	−0.012 (0.010)	−0.016 (0.018)	−0.015 (0.015)	−0.010 (0.008)	0.001 (0.007)	0.004 (0.011)	−0.000 (0.008)	−0.003 (0.008)	−0.001 (0.008)	0.001 (0.007)	0.000 (0.008)	−0.006 (0.008)
Inflation	0.024** (0.011)	0.026** (0.011)	0.024** (0.011)	−0.012 (0.024)	0.023** (0.011)	0.024** (0.011)	−0.020 (0.023)	0.024** (0.011)	0.025** (0.011)	0.022** (0.011)	0.024** (0.011)	0.024** (0.011)	0.022** (0.011)
Broad Money Growth	0.052 (0.043)	0.060 (0.044)	0.055 (0.044)	0.102 (0.071)	0.053 (0.044)	0.055 (0.044)	0.107 (0.070)	0.054 (0.044)	0.055 (0.044)	0.053 (0.043)	0.053 (0.043)	0.050 (0.043)	0.056 (0.044)
Unemployment Rate	0.114*** (0.029)	0.101*** (0.029)	0.105*** (0.029)	0.233*** (0.055)	0.110*** (0.029)	0.108*** (0.029)	0.248*** (0.055)	0.108*** (0.029)	0.108*** (0.029)	0.125*** (0.033)	0.113*** (0.030)	0.111*** (0.030)	0.131*** (0.030)
World Uncertainty Index	−0.003 (0.021)	−0.001 (0.022)	−0.001 (0.022)	−0.012 (0.034)	−0.015 (0.022)	−0.003 (0.021)	−0.011 (0.033)	−0.005 (0.022)	−0.004 (0.021)	−0.003 (0.021)	−0.004 (0.021)	−0.005 (0.022)	0.002 (0.022)
Country Risk Premium	−0.018 (0.014)	−0.014 (0.014)	−0.014 (0.014)	−0.053* (0.028)	−0.012 (0.014)	−0.014 (0.014)	−0.041 (0.028)	−0.016 (0.014)	−0.017 (0.014)	−0.015 (0.014)	−0.015 (0.014)	−0.016 (0.014)	−0.017 (0.014)
Exchange Rate Change	0.113 (0.148)	0.098 (0.149)	0.114 (0.148)	−0.006 (0.241)	0.247 (0.153)	0.093 (0.153)	0.068 (0.242)	0.109 (0.151)	0.135 (0.146)	0.146 (0.149)	0.100 (0.150)	0.127 (0.148)	0.106 (0.152)
BTC return	0.023 (0.015)	0.017** (0.008)	0.018 (0.016)	0.026** (0.013)	−0.028*** (0.008)	−0.003 (0.004)	−0.008 (0.005)	0.005 (0.009)	0.015** (0.006)	0.019 (0.015)	0.011 (0.013)	0.003 (0.005)	0.028*** (0.007)
Intercept	−2.054 (1.626)	−0.763 (1.443)	−0.802 (1.439)	−0.803 (2.283)	0.879 (1.509)	−0.378 (1.596)	−0.550 (2.348)	−0.980 (1.442)	−1.796 (1.449)	−2.468 (1.949)	−1.520 (1.661)	−1.055 (1.463)	−2.204 (1.507)
Number of observations	393	393	393	193	393	393	193	393	393	393	393	393	393
McFadden pseudo R-squared	27.4	27.7	27.2	36.5	29.5	27.1	35.9	27.0	28.1	27.3	27.1	27.1	30.1

Estimation of the probit model within the first stage of the Heckman's procedure. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %.

Table 10
Heckman's second stage - OLS regression.

Dependent Variable: Y1 (Total Localbitcoins volume by country-year)														
Overall Score	−2.250*** (0.675)													
Property Rights		−0.546 (0.381)												
Government Integrity			−1.656 (1.017)											
Judicial Effectiveness				0.337 (0.303)										
Tax Burden					−0.009 (0.368)									
Government Spending						−0.172 (0.145)								
Fiscal Health							−0.118 (0.115)							
Business Freedom								−0.208 (0.307)						
Labor Freedom									0.227 (0.442)					
Monetary Freedom										−3.743*** (0.991)				
Trade Freedom											−0.855** (0.356)			
Investment Freedom												−0.565** (0.261)		
Financial Freedom													−1.264*** (0.406)	
Corruption Index	−0.079 (0.251)	−0.153 (0.309)	1.077 (1.007)	−0.864** (0.354)	−0.613** (0.297)	−0.686** (0.296)	−0.583** (0.281)	−0.542* (0.283)	−0.570* (0.296)	0.167 (0.248)	−0.533* (0.275)	−0.365 (0.243)	−0.188 (0.241)	
Average Restrictions PABR	−49.156*** (14.522)	−46.506*** (14.038)	−47.793*** (15.506)	−27.860** (13.773)	−58.870*** (18.836)	−48.736*** (15.206)	−22.254* (12.616)	−51.120*** (16.010)	−51.029*** (16.270)	−31.272*** (10.755)	−47.673*** (14.845)	−43.937*** (14.125)	−44.387*** (12.854)	
Average Restrictions SIAR	10.926 (9.659)	17.305* (10.270)	22.473** (10.844)	18.266* (11.051)	6.428 (12.290)	15.733 (10.604)	26.946* (14.019)	15.537 (10.561)	14.677 (10.095)	19.371** (9.115)	14.773 (9.691)	6.247 (9.880)	1.038 (10.547)	
GDP per Capita	21.118*** (7.842)	19.024** (7.661)	20.345** (8.391)	9.653* (5.252)	18.145** (7.316)	17.456** (7.718)	11.969** (5.707)	19.650** (8.405)	17.066** (6.704)	9.806* (5.393)	21.169*** (8.021)	17.772** (7.074)	18.134** (7.258)	
GDP Growth	−1.245 (1.197)	−1.337 (1.238)	−1.293 (1.239)	−4.104 (2.571)	−0.791 (1.257)	−1.048 (1.209)	−3.113 (2.493)	−1.327 (1.256)	−1.338 (1.263)	−1.319 (1.270)	−1.260 (1.214)	−0.839 (1.202)	−1.745 (1.218)	
Inflation	−0.566 (1.254)	−0.088 (1.295)	0.207 (1.365)	−2.078 (2.561)	−1.858 (1.718)	−0.285 (1.369)	0.592 (3.053)	−0.762 (1.386)	−1.475 (1.554)	0.540 (1.177)	−0.574 (1.191)	−0.780 (1.302)	−1.430 (1.261)	
Broad Money Growth	−0.794 (0.596)	−0.935 (0.659)	−0.849 (0.623)	1.387 (1.109)	−0.735 (0.579)	−0.803 (0.602)	0.810 (1.070)	−0.793 (0.590)	−0.670 (0.566)	−1.357** (0.667)	−0.883 (0.594)	−0.754 (0.578)	−0.745 (0.571)	
Unemployment Rate	0.537 (0.471)	0.931* (0.490)	0.757* (0.442)	−0.621 (0.468)	0.729 (0.537)	0.728 (0.483)	−0.766 (0.467)	0.922* (0.498)	1.120* (0.626)	0.303 (0.390)	0.807* (0.484)	1.099** (0.541)	0.921* (0.516)	
World Uncertainty Index	−0.786* (0.428)	−0.619 (0.407)	−0.742* (0.411)	−1.135* (0.596)	−0.727* (0.410)	−0.837* (0.434)	−0.891 (0.601)	−0.830* (0.436)	−0.820** (0.418)	−0.903** (0.433)	−0.828* (0.435)	−0.743* (0.415)	−0.899** (0.443)	
Country Risk Premium	4.117 (3.777)	4.457 (3.868)	3.957 (4.057)	4.353 (4.134)	12.087* (6.676)	5.256 (4.537)	−0.261 (5.105)	7.059 (4.603)	8.784* (5.249)	−5.681 (3.843)	6.015 (3.978)	4.858 (4.262)	5.404 (3.652)	
Exchange Rate Change	−0.862** (0.402)	−0.899** (0.434)	−0.838** (0.427)	−0.131 (0.474)	−1.209** (0.573)	−0.906** (0.437)	−0.419 (0.574)	−0.919** (0.443)	−0.947** (0.457)	0.141 (0.302)	−0.833** (0.416)	−0.728* (0.389)	−0.810** (0.397)	
BTC return	−1.892** (0.940)	−1.832** (0.921)	−2.015** (0.957)	−1.129* (0.655)	−2.410** (1.121)	−2.105** (1.006)	−0.760 (0.715)	−2.150** (1.008)	−2.208** (1.024)	−1.148 (0.760)	−2.056** (0.990)	−1.895** (0.933)	−2.065** (0.987)	
Lambda	−44.625* (23.439)	−40.757* (23.272)	−36.458 (25.063)	−37.036 (24.043)	−87.757** (41.010)	−45.320* (26.484)	−15.834 (29.793)	−52.865* (27.238)	−61.879** (29.793)	12.754 (22.863)	−46.224** (23.581)	−39.212 (25.565)	−52.772** (22.646)	
Intercept	49.327 (47.862)	−59.498 (46.968)	−82.377 (56.444)	12.579 (44.851)	−36.986 (48.661)	−30.349 (54.880)	−15.986 (48.256)	−52.721 (47.520)	−56.937 (56.456)	264.166*** (74.256)	−15.982 (46.972)	−31.074 (40.714)	14.005 (40.831)	
Number of observations	238	238	238	118	238	238	118	238	238	238	238	238	238	
McFadden pseudo R-squared	5.0	3.5	4.1	8.7	5.9	3.7	5.9	3.5	4.7	10.0	3.7	4.3	4.6	

Estimation of the linear model augmented with the Lambda statistics. Lambda is derived from the first stage of the Heckman's procedure. ***, **, * indicates statistical significance at 1 %, 5 %, and 10 %.

Table 11
Regressions with different scores.

Dependent Variable: Y1 (Total Localbitcoins volume by country-year)			
Fragile State Index	1.188*** (0.439)		
Political Stability		−33.420*** (10.403)	
Regulatory Quality			−45.543** (20.949)
Corruption Index	0.244 (0.357)	−0.014 (0.269)	0.706 (0.530)
Average Restrictions PABR	−33.213*** (11.840)	−41.588*** (11.071)	−36.525*** (10.019)
Average Restrictions SIAR	34.453*** (12.660)	26.209*** (8.834)	24.223*** (8.068)
GDP per Capita	24.502*** (9.318)	26.514*** (9.497)	25.154*** (9.666)
GDP Growth	−2.864 (2.046)	−0.264 (1.339)	−1.263 (1.203)
Inflation	1.129 (1.228)	0.189 (0.630)	0.998 (0.664)
Broad Money Growth	−0.376 (0.703)	−1.289* (0.698)	−1.126* (0.665)
Unemployment Rate	−0.353 (0.499)	1.171** (0.529)	0.396 (0.410)
World Uncertainty Index	−0.970** (0.474)	−0.840* (0.433)	−1.020** (0.481)
Country Risk Premium	−1.538 (1.323)	−3.195*** (0.999)	−2.247** (1.010)
Exchange Rate Change	−0.245 (0.429)	−0.284 (0.323)	−0.497 (0.345)
BTC return	−1.136* (0.678)	−1.581* (0.855)	−1.807** (0.908)
Intercept	−242.908** (98.462)	−179.712** (72.084)	−169.149** (76.278)
Number of observations	154	238	238
Adjusted R-square	5.8	7.8	6.4

This table reports regression results where the dependent variable Y1 is regressed on the Fragile State Index, the Political Stability and Regulatory Quality indicators, and a set of control variables. Y1 is the annual volume of BTC traded on Localbitcoins, scaled by one thousand. See [Subsection 3.1](#) for more details. Standard errors, in parentheses, follow coefficient estimates. Standard errors are corrected for heteroscedasticity following [White \(1980\)](#). ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

$$\hat{\theta}_{ij} = \hat{\alpha}_0 + \sum_z \hat{\alpha}_z \times \text{EXPLANATORY_VARIABLES}_{z,ij}$$

Lambda is calculated for each country-year observation of our sample and introduced as an additional explanatory variable in the linear model presented in [Subsection 3.3](#). This is the second stage of the Heckman's procedure. The augmented panel regression, estimated with OLS, is exhibited in [Table 10](#).

The significance of the lambda statistic in multiple regressions indicates the presence of sample selection bias. By correcting this bias, the study ensures that the estimated relationship between BTC trade volume and institutional features is not driven by non-random sample selection, thereby addressing another potential source of endogeneity. In addition, the probit regressions show a statistically significant coefficient associated with the corruption index. This result is consistent with [Asthana \(2024\)](#), who finds cross-country evidence that corruption is important in explaining crypto adoption rates. Our study confirms this result, as corruption is positively related to the likelihood of being covered by Localbitcoins in our sample. Nevertheless, our results also show that the severity of corruption poorly explains the traded volume on Localbitcoins in the cross-section of countries, conditionally on being covered by Localbitcoins.

6.2. Political stability and fragile state index

Running regressions using Political Stability, Fragile State, and Regulatory Quality Indexes ([Table 11](#)) instead of Heritage's Overall

Economic Freedom score allows the study to account for the broader institutional environment's role in BTC trading activity. Moreover, those three indexes are similar to Heritage's Economic Freedom score. Thus, regressing them serves as a test for the robustness of our study. In our case, we obtain significant results with similar scores from different sources and using different methodologies. This confirms our main analysis: whenever institutions are deficient, or whenever the perception of those institutions is negative, BTC trading volume in DEXs is higher.

6.3. Early and late adopters

We redo the regressions in two different periods, 2013–2016 and 2017–2019, to separate early adopters of Localbitcoins and Paxful from late adopters. The results in [Tables 12 and 13](#) show no significant differences concerning the main results ([Table 5](#)).¹²

The distinction between early and late adopters of Localbitcoins and Paxful, analyzed through separate regressions for the aforementioned periods, aims to identify if there are time-period-specific effects that might influence the results. Suppose institutional features endogenously influenced BTC trade volumes. In that case, one might expect to see significant differences in the determinants of trade volume between early and late adopters due to changes in institutional quality over time. However, the results showing no significant differences between these periods ([Tables 12 and 13](#)) suggest that the relationship between BTC trade volume and institutional features is consistent over time. Indeed, the coefficients remain significant while keeping the same sign, indicating the same effect in both subperiods. Interestingly, the unemployment rate is relatively more substantial during the 2013–2016 subperiod, which suggests that the « urgency » of trading was relatively stronger in those countries with higher unemployment when DEXs emerged. Something less likely for late adopters.

7. Policy implications and conclusion

Our findings highlight the relationship between economic governance, capital restrictions, and macroeconomic control variables with BTC transaction volumes in DEXs. Although our results are comparable to those of [Alnasaa et al. \(2022\)](#), our conclusions differ. Indeed, since we had access to a country-specific dataset on transaction volumes, we did not need to resort to qualitative survey-based data. Therefore, our results suggest that users are more likely to turn to DEXs when facing threats to overall economic freedoms, poor monetary governance, and heavy tax burdens. Additionally, capital restrictions influence the transaction volumes in DEXs differently depending on whether they are related to purchasing or selling assets abroad.

Consequently, our findings imply some recommendations for practitioners and regulators. Private investors should be aware that increased restrictions on sales and issues of domestic assets can lead to higher demand for alternative assets like BTC, which may impact market dynamics and investment opportunities. Moreover, investors should monitor and consider the impact of economic governance and capital restrictions on crypto-asset markets when making investment decisions. These factors may affect the potential returns and risks associated with such investments.

Institutional investors like banks and investment funds should acknowledge the growing interest in DEX platforms and crypto-assets, as these may provide opportunities for portfolio diversification and risk management. Traditional financial institutions may face increased competition from DEXs and other blockchain-based financial services, which could challenge their market dominance and revenue streams. Crypto-asset transactions on DEXs can be difficult to trace, posing

¹² The two variables Fiscal Health and Judicial Effectiveness from [Heritage.org](#) are absent in the regressions since they do not exist for the period 2013–2016.

Table 12

Annual Localbitcoins data regression for the period 2013–2016 only.

Dependent Variable: Y1 (Total Localbitcoins volume by country-year)											
Overall Score	−1.619*										
	(0.914)										
Property Rights		0.639									
		(0.599)									
Government Integrity			−5.768**								
			(2.552)								
Tax Burden				−1.499**							
				(0.724)							
Government Spending					−0.377						
					(0.288)						
Business Freedom						−0.150					
						(0.500)					
Labor Freedom							1.196				
							(0.839)				
Monetary Freedom								−5.762***			
								(1.721)			
Trade Freedom									−0.378		
									(0.725)		
Investment Freedom										−0.561	
										(0.457)	
Financial Freedom											−0.461
											(0.511)
Corruption Index	−0.487	−1.590*	4.915**	−1.673**	−1.115	−0.855	−0.957	0.299	−0.857	−0.612	−0.691
	(0.586)	(0.937)	(2.285)	(0.841)	(0.681)	(0.615)	(0.652)	(0.422)	(0.621)	(0.517)	(0.554)
Average Restrictions PABR	−53.031***	−59.247***	−56.575***	−54.745***	−53.495***	−55.891***	−47.888***	−49.858***	−54.448***	−51.656***	−52.587***
	(16.614)	(18.839)	(17.880)	(17.199)	(16.820)	(18.085)	(15.309)	(14.811)	(17.447)	(15.945)	(16.172)
Average Restrictions SLAR	30.835**	46.750***	32.841**	38.967***	36.527**	40.021**	38.893**	20.452	37.944**	25.078*	35.154**
	(13.759)	(16.793)	(14.813)	(14.912)	(14.990)	(16.409)	(15.743)	(13.644)	(14.846)	(14.793)	(13.790)
GDP per Capita	30.836*	34.401**	32.209*	39.187**	29.631*	31.557*	25.005*	15.730	31.766**	27.532*	29.038*
	(16.004)	(16.959)	(16.442)	(18.675)	(15.950)	(16.934)	(13.185)	(12.190)	(15.916)	(14.441)	(15.261)
GDP Growth	−3.922	−3.048	−4.429	−2.895	−3.354	−3.852	−4.020	−4.901	−3.751	−3.751	−4.245
	(2.805)	(2.651)	(2.888)	(2.748)	(2.726)	(2.760)	(2.805)	(3.316)	(2.759)	(2.658)	(2.990)
Inflation	2.419	2.771	3.026	3.355	3.035	2.550	2.029	0.843	2.506	1.759	2.366
	(1.962)	(2.012)	(2.186)	(2.228)	(2.159)	(2.028)	(1.648)	(1.631)	(2.077)	(1.591)	(1.882)
Broad Money Growth	−1.023	−0.906	−0.812	−0.987	−0.911	−1.040	−1.189	−1.766**	−1.101	−1.034	−1.027
	(0.793)	(0.833)	(0.827)	(0.896)	(0.811)	(0.803)	(0.887)	(0.813)	(0.798)	(0.775)	(0.785)
Unemployment Rate	1.768**	2.066**	2.051**	1.154	1.650**	2.012**	2.591**	0.947	2.024**	2.347**	2.073**
	(0.840)	(0.923)	(0.932)	(0.734)	(0.763)	(0.897)	(1.231)	(0.697)	(0.895)	(1.057)	(0.923)
World Uncertainty Index	−4.345	−4.188	−1.418	−5.115	−4.705	−4.230	−3.221	−2.538	−4.037	−3.894	−4.349
	(8.997)	(9.040)	(8.141)	(9.043)	(9.106)	(9.098)	(8.471)	(8.384)	(9.262)	(8.905)	(9.046)
Country Risk Premium	−2.199	−0.041	−2.588	−0.636	−2.034	−1.000	0.236	−3.664*	−0.933	−1.276	−1.888
	(1.874)	(1.942)	(1.933)	(1.965)	(1.985)	(1.777)	(1.887)	(1.953)	(1.761)	(1.635)	(1.993)
Exchange Rate Change	−1.848	−1.910	−1.851	−2.150	−2.049	−1.840	−1.806	−1.036	−1.787	−1.721	−1.856
	(1.497)	(1.506)	(1.458)	(1.575)	(1.547)	(1.499)	(1.432)	(1.304)	(1.550)	(1.425)	(1.494)
BTC return	−13.642	−13.487	−12.180	−15.005	−14.319	−13.367	−12.018	−11.181	−13.095	−12.841	−13.578
	(16.417)	(16.467)	(15.715)	(16.526)	(16.609)	(16.587)	(15.454)	(15.422)	(16.827)	(16.204)	(16.483)
Intercept	28.765	−109.831	−148.478	31.223	−6.871	−66.714	−122.182	441.764*	−54.180	−21.576	−27.466
	(208.297)	(210.430)	(191.150)	(217.280)	(218.403)	(209.502)	(193.763)	(265.282)	(197.858)	(210.493)	(213.658)
Number of observations	120	120	120	120	120	120	120	120	120	120	120
Adjusted R-square	4.0	3.6	7.3	5.9	3.8	3.3	5.7	10.9	3.4	3.8	3.6

This table reports regression results where the dependent variable Y1 is regressed on Heritage indexes and a set of control variables. Y1 is the annual volume of BTC traded on Localbitcoins, scaled by one thousand. See Subsection 3.1 for more details. These regressions include the period 2013–2016 only. Standard errors, in parentheses, follow coefficient estimates. Standard errors are corrected for heteroscedasticity following White (1980). ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

Table 13
Annual Localbitcoins data regression for the period 2017–2019 only.

Dependent Variable: Y1 (Total Localbitcoins volume by country-year)											
Overall Score	−1.898*** (0.636)										
Property Rights		−1.160** (0.480)									
Government Integrity			0.199 (0.495)								
Tax Burden				−0.462** (0.216)							
Government Spending					−0.167 (0.130)						
Business Freedom						−0.221 (0.442)					
Labor Freedom							0.038 (0.279)				
Monetary Freedom								−2.428*** (0.698)			
Trade Freedom									−1.317*** (0.473)		
Investment Freedom										−0.569** (0.252)	
Financial Freedom											−0.896*** (0.307)
Corruption Index	−0.091 (0.250)	0.039 (0.243)	−0.703 (0.692)	−0.662** (0.292)	−0.548* (0.284)	−0.443 (0.308)	−0.501* (0.266)	−0.140 (0.201)	−0.396 (0.260)	−0.322 (0.213)	−0.102 (0.216)
Average Restrictions PABR	−22.968** (9.454)	−21.979** (9.207)	−18.037** (8.686)	−21.243** (9.558)	−20.882** (9.989)	−18.744** (9.185)	−18.285* (10.576)	−20.737*** (7.940)	−16.740* (8.979)	−16.500** (8.221)	−15.147* (8.020)
Average Restrictions SIAR	29.529** (11.909)	32.619** (12.733)	28.323** (11.799)	30.087** (12.733)	30.490** (13.016)	31.150** (12.946)	29.820** (12.598)	13.624* (8.167)	29.242** (12.322)	18.862* (10.322)	19.038* (9.879)
GDP per Capita	12.203** (5.577)	14.883** (6.660)	9.776* (5.238)	10.247* (5.519)	7.704 (5.303)	11.419* (6.412)	9.848* (5.593)	2.672 (3.619)	13.205** (5.891)	10.121** (4.831)	7.450* (4.374)
GDP Growth	−2.755 (2.138)	−3.199 (2.214)	−2.590 (2.238)	−2.529 (2.225)	−2.743 (2.308)	−2.930 (2.085)	−2.689 (2.220)	−1.513 (1.908)	−2.789 (2.235)	−2.539 (2.088)	−3.197 (2.066)
Inflation	2.544* (1.435)	1.607 (1.342)	1.918 (1.427)	2.267 (1.441)	2.148 (1.461)	2.092 (1.450)	1.941 (1.611)	−0.159 (1.312)	1.931 (1.373)	1.340 (1.368)	1.622 (1.289)
Broad Money Growth	0.394 (0.799)	0.372 (0.851)	0.462 (0.908)	0.366 (0.872)	0.402 (0.866)	0.417 (0.915)	0.486 (0.868)	0.674 (0.738)	0.738 (0.867)	0.812 (0.845)	0.725 (0.785)
Unemployment Rate	−1.121** (0.513)	−0.434 (0.393)	−0.609 (0.421)	−0.995** (0.487)	−0.751 (0.478)	−0.746* (0.433)	−0.639 (0.474)	−0.371 (0.355)	−0.987** (0.494)	−0.615 (0.439)	−0.666 (0.432)
World Uncertainty Index	−0.723 (0.459)	−0.807* (0.488)	−0.747 (0.491)	−0.755 (0.474)	−0.743 (0.482)	−0.760 (0.487)	−0.768 (0.505)	−1.041** (0.511)	−0.817* (0.487)	−0.757 (0.476)	−0.797* (0.472)
Country Risk Premium	−5.590** (2.176)	−4.041** (1.819)	−2.615 (1.690)	−2.920* (1.710)	−3.626* (1.989)	−2.822* (1.702)	−2.750 (1.824)	−5.918*** (2.229)	−3.576** (1.684)	−2.781* (1.611)	−4.383** (1.774)
Exchange Rate Change	−0.614 (0.435)	−0.483 (0.415)	−0.514 (0.438)	−0.556 (0.438)	−0.506 (0.440)	−0.534 (0.443)	−0.505 (0.494)	0.037 (0.366)	−0.638 (0.431)	−0.379 (0.418)	−0.495 (0.406)
BTC return	−0.377 (0.489)	−0.638 (0.542)	−0.605 (0.550)	−0.626 (0.545)	−0.548 (0.549)	−0.602 (0.542)	−0.613 (0.558)	−0.417 (0.500)	−0.548 (0.527)	−0.518 (0.527)	−0.501 (0.512)
Intercept	71.882 (48.398)	−13.882 (40.606)	−21.951 (41.208)	20.636 (47.319)	14.311 (51.074)	−22.632 (42.126)	−24.414 (42.327)	238.003*** (73.041)	50.943 (45.624)	4.887 (35.931)	41.859 (38.870)
Number of observations	118	118	118	118	118	118	118	118	118	118	118
Adjusted R-square	12.9	10.6	5.1	7.2	5.7	5.2	5.0	19.0	9.0	11.3	14.9

This table reports regression results where the dependent variable Y1 is regressed on Heritage indexes and a set of control variables. Y1 is the annual volume of BTC traded on Localbitcoins, scaled by one thousand. See [Subsection 3.1](#) for more details. These regressions include the period 2017–2019 only. Standard errors, in parentheses, follow coefficient estimates. Standard errors are corrected for heteroscedasticity following [White \(1980\)](#). ***, **, and * indicate significance at 1 %, 5 %, and 10 %, respectively.

potential challenges for banks and investment funds in complying with AML and KYC regulations. Therefore, institutional investors should consider exploring opportunities in the blockchain and crypto-asset space, including offering crypto-asset-related products and services to their clients, such as crypto custody, trading, and investment solutions (such as the recent approval of BTC and ETH ETFs by the SEC), alongside their more traditional financial products. Moreover, collaboration with regulators and other stakeholders is crucial for banks and investment funds to establish industry standards and best practices for dealing with the risks associated with DEXs and crypto-assets, such as fraud, market manipulation, and cyber threats.

Concerning regulators, these agents should carefully consider the potential consequences of implementing heavy tax burdens and loose monetary governance, as they may incentivize users to turn to unregulated platforms like DEXs. Furthermore, restrictions on the holding and transactions of crypto-assets (and even more so, outright bans) can also incentivize users to fly to unregulated DEXs. This happened in several countries of our sample: China, Saudi Arabia, Pakistan, Egypt, Morocco, Colombia, Indonesia, Iran, Russia, Turkey, and Nigeria. All banned or implemented strict restrictions on the holding, selling, trading, mining, and issuing of crypto-assets. Our results suggest that people use DEX precisely to circumvent restrictive regulations and poor institutions. In this regard, the case of Nigeria is enlightening. In 2017, the Central Bank of Nigeria banned Nigerian financial institutions from doing crypto-related business, a ban that was lifted in December 2023. Meanwhile, Nigeria became one of the largest crypto-market in the world, especially for peer-to-peer transactions on unregulated decentralized platforms (Chainalysis, 2024). An outright ban on decentralized technology is doomed to fail because people can easily circumvent the ban. A policy implication is that rather than wasting resources to try to ban unbanable technology, financial authorities should instead devote resources to nurture a sound institutional framework for crypto-assets with relevant and supportive regulation, with straightforward, easy-to-implement and enforce rules to deter illicit activities, support innovation, protect customers, maintain financial stability, and collect taxes.

To maintain stability in the crypto-asset markets and protect investors, regulators should monitor the relationship between economic governance, capital restrictions, and the use of DEXs. Striking a balance between promoting economic freedom and ensuring proper oversight is crucial. In this manner, regulators in institutionally strong countries may significantly influence the global regulatory landscape for crypto-assets, setting examples and guidelines for other nations to follow. Indeed, they typically have more resources and expertise available to address challenges related to crypto-asset regulation, allowing them to develop more comprehensive and effective policies. Regulators from institutionally strong countries can thus encourage collaboration with regulators from

institutionally weak countries to share best practices, resources, and knowledge, fostering a more consistent and coordinated global approach to crypto-asset regulation. These measures can help balance protecting investors, preventing illicit activities, and fostering a competitive environment that promotes innovation in the crypto-asset and blockchain sectors.

Concomitantly, regulators in institutionally weak countries may face more significant challenges in developing and enforcing crypto-asset regulations, including limited resources, expertise, and a potentially weaker legal framework. The potential for higher corruption levels and weaker governance in institutionally weak countries may lead to increased use of DEXs and crypto-assets to evade poor institutional frameworks, making regulation even more critical. Indeed, ill-designed and poorly implemented regulatory frameworks may imply high platform compliance costs and user administrative burdens, leading to detrimental consequences. For instance, agents unable (or unwilling) to comply with regulation may want to keep trading BTC on DEXs by using more complex obfuscation tools and protocols, such as mixers and peer-to-peer networks running on the TOR network. This flight toward onion routing networks and obfuscation tools would be at odds with regulators' initial purposes and with the technology-neutral approach of regulation and the 'same activity, same risk, same regulation' principle as advocated by international regulatory authorities (such as the Financial Stability Board, the FATF, and the IMF). Therefore, regulators in institutionally weak countries should prioritize building capacity and expertise in the crypto-asset and blockchain sectors. They could thus seek partnerships and collaborations with regulators from institutionally strong countries, international organizations, and industry stakeholders to access resources, expertise, and support in developing and implementing effective regulatory frameworks.

CRedit authorship contribution statement

Gabriel A. Giménez Roche: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Investigation, Conceptualization. **Antoine Noël:** Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Loïc Sauce:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization.

Declaration of competing interest

The authors have NO affiliations with or involvement in any organization or entity with any financial interest, or non-financial interest in the subject matter or materials discussed in this manuscript.

Appendix A. Country and year fixed and random effects

Table 14

Fixed and random effects results for governance variables.

	Country F.E.		Year F.E.		Country R.E.		Year R.E.	
	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat
Dependent Variable 1. Localbitcoins Volume								
Overall Score	−2.060	−1.064	−1.136	−1.694	−1.530	−1.341	−1.341	−1.787
Property Rights	0.022	0.052	0.172	0.431	0.087	0.223	−0.145	−0.357
Government Integrity	−1.315	−1.893	−0.896	−1.144	−1.463	−2.031	−0.836	−1.009
Tax Burden	−5.889	−3.759	−0.678	−2.000	−0.995	−1.567	−0.713	−1.874
Government Spending	−0.993	−1.371	−0.145	−0.756	−0.204	−0.578	−0.139	−0.650
Business Freedom	−0.247	−0.318	0.231	0.557	0.177	0.281	0.238	0.497
Labor Freedom	1.933	2.155	0.608	2.220	0.835	1.705	0.640	2.083
Monetary Freedom	−1.444	−1.489	−2.821	−4.350	−2.174	−2.733	−2.889	−4.121

(continued on next page)

Table 14 (continued)

	Country F.E.		Year F.E.		Country R.E.		Year R.E.	
	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat
Trade Freedom	0.721	0.658	−0.566	−0.958	−0.065	−0.078	−0.616	−0.959
Investment Freedom	−3.530	−4.300	−0.403	−1.790	−1.062	−2.614	−0.458	−1.837
Financial Freedom	−3.543	−3.578	−0.543	−1.816	−1.402	−2.656	−0.603	−1.813
Dependent Variable 2. Localbitcoins Volume Scaled by Outstanding BTC								
Overall Score	−0.148	−1.118	−0.073	−1.604	−0.102	−1.326	−0.087	−1.714
Property Rights	0.000	0.009	0.015	0.551	0.005	0.193	−0.009	−0.338
Government Integrity	−0.095	−1.996	−0.065	−1.221	−0.104	−2.117	−0.064	−1.136
Tax Burden	−0.412	−3.841	−0.047	−2.021	−0.067	−1.567	−0.048	−1.849
Government Spending	−0.077	−1.556	−0.010	−0.792	−0.015	−0.626	−0.009	−0.642
Business Freedom	−0.010	−0.182	0.017	0.601	0.016	0.367	0.016	0.505
Labor Freedom	0.146	2.373	0.043	2.280	0.060	1.814	0.044	2.139
Monetary Freedom	−0.106	−1.598	−0.190	−4.298	−0.150	−2.777	−0.192	−4.048
Trade Freedom	0.047	0.628	−0.035	−0.859	−0.005	−0.085	−0.039	−0.904
Investment Freedom	−0.245	−4.363	−0.025	−1.627	−0.070	−2.550	−0.029	−1.724
Financial Freedom	−0.238	−3.515	−0.033	−1.638	−0.090	−2.538	−0.038	−1.670
Dependent Variable 3. Localbitcoins and Paxful Volumes								
Overall Score	−2.117	−1.136	−1.238	−1.704	−1.750	−1.382	−1.456	−1.790
Property Rights	−0.007	−0.016	0.106	0.244	0.078	0.203	−0.203	−0.462
Government Integrity	−1.172	−1.749	−0.837	−0.985	−1.384	−1.97	−0.752	−0.836
Tax Burden	−5.661	−3.749	−0.75	−2.041	−1.260	−1.712	−0.796	−1.932
Government Spending	−0.814	−1.165	−0.139	−0.669	−0.209	−0.519	−0.132	−0.567
Business Freedom	−0.329	−0.440	0.157	0.348	0.027	0.041	0.154	0.296
Labor Freedom	1.596	1.842	0.739	2.493	0.906	1.651	0.779	2.344
Monetary Freedom	−1.339	−1.433	−3.027	−4.303	−2.066	−2.507	−3.096	−4.070
Trade Freedom	0.885	0.839	−0.637	−0.996	0.131	0.148	−0.676	−0.971
Investment Freedom	−3.223	−4.061	−0.39	−1.598	−1.282	−2.774	−0.441	−1.629
Financial Freedom	−3.419	−3.583	−0.523	−1.612	−1.700	−2.866	−0.578	−1.601
Dependent Variable 4. Localbitcoins and Paxful Volumes Scaled by Outstanding BTC								
Overall Score	−0.153	−1.198	−0.08	−1.625	−0.117	−1.382	−0.094	−1.727
Property Rights	−0.001	−0.050	0.011	0.384	0.005	0.181	−0.013	−0.431
Government Integrity	−0.087	−1.893	−0.062	−1.092	−0.101	−2.093	−0.06	−0.988
Tax Burden	−0.404	−3.909	−0.051	−2.064	−0.084	−1.721	−0.053	−1.911
Government Spending	−0.066	−1.386	−0.010	−0.713	−0.016	−0.592	−0.009	−0.569
Business Freedom	−0.015	−0.289	0.013	0.414	0.007	0.166	0.011	0.326
Labor Freedom	0.126	2.126	0.050	2.524	0.066	1.799	0.053	2.373
Monetary Freedom	−0.099	−1.551	−0.203	−4.273	−0.145	−2.593	−0.205	−4.017
Trade Freedom	0.057	0.793	−0.039	−0.899	0.007	0.113	−0.043	−0.918
Investment Freedom	−0.229	−4.223	−0.024	−1.480	−0.085	−2.752	−0.028	−1.556
Financial Freedom	−0.234	−3.580	−0.032	−1.483	−0.110	−2.776	−0.036	−1.504

Coeff and T-stat stand for regression coefficients and t-statistics, respectively. F.E. and R.E stand for fixed and random effects, respectively. List of control variables: Corruption index, real GDP per capital (log), GDP growth, inflation, money growth, employment rate, world uncertainty index, exchange rate change, lagged Bitcoin return.

Data availability

Data will be made available on request.

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